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Lipid nanoparticle (LNP) mediated mRNA delivery in neurodegenerative diseases

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ABSTRACT

Neurodegenerative diseases (NDD) are characterized by the progressive loss of neurons and the impairment of cellular functions. Messenger RNA (mRNA) has emerged as a promising therapy for treating NDD, as it can encode missing or dysfunctional proteins and anti-inflammatory cytokines or neuroprotective proteins to halt the progression of these diseases. However, effective mRNA delivery to the central nervous system (CNS) remains a significant challenge due to the limited penetration of the blood-brain barrier (BBB). Lipid nanoparticles (LNPs) offer an efficient solution by encapsulating and protecting mRNA, facilitating transfection and intracellular delivery. This review discusses the pathophysiological mechanisms of neurological disorders, including Parkinson's disease (PD), Alzheimer's disease (AD), multiple sclerosis (MS), Huntington's disease (HD), ischemic stroke, spinal cord injury, and Friedreich's ataxia. Additionally, it explores the potential of LNP-mediated mRNA delivery as a therapeutic strategy for these diseases. Various approaches to overcoming BBB-related challenges and enhancing the delivery and efficacy of mRNA-LNPs are discussed, including non-invasive methods with strong potential for clinical translation. With advancements in artificial intelligence (AI)-guided mRNA and LNP design, targeted delivery, gene editing, and CAR-T cell therapy, mRNA-LNPs could significantly transform the treatment landscape for NDD, paving the way for future clinical applications.

1. Introduction

Neurodegenerative diseases (NDD), including Alzheimer's disease (AD) [1], Parkinson's disease (PD) [2], multiple sclerosis (MS) [3], and stroke [4], are leading causes of physical and cognitive disability worldwide [5]. The World Health Organization (WHO) predicts NDD will soon surpass cancer as the second leading cause of death, after cardiovascular diseases [5]. The prevalence of NDD is expected to double in the next two decades, yet limited understanding and lack of effective treatments remain major challenges. Future research should focus on NDD mechanisms, prevention, early diagnosis, effective treatment strategies, and improving long-term patient care [6,7].

Current treatment strategies for NDD primarily focus on symptom management through pharmacological and non-pharmacological interventions [7]. Among emerging approaches, RNA-based therapeutics, particularly messenger RNA (mRNA) therapy, show great promise by enabling targeted protein synthesis and identifying disease-related genes [8]. Synthetic mRNA allows efficient exogenous protein expression, rapid gene translation, and scalable production [5,9]. Compared to protein or other nucleic acid (NA)-based therapies, mRNA offers advantages such as controlled protein production, no risk of insertional mutagenesis, and functionality without nuclear entry [10]. Additionally, modifications to mRNA can enhance protein production while minimizing inflammatory responses [11,12].

A major challenge in mRNA-based therapy is ensuring that fragile mRNA molecules reach target cells intact without degradation [13]. Nanocarriers are crucial for protecting mRNA and enabling efficient delivery. Ideal carriers should encapsulate mRNA securely, maintain stability in circulation to prevent degradation by nucleases, and facilitate targeted cellular uptake and intracellular release. Additionally, they must be biocompatible, cost-effective, and easy to produce [14]. mRNA vectors are classified as viral and non-viral, with the latter offering

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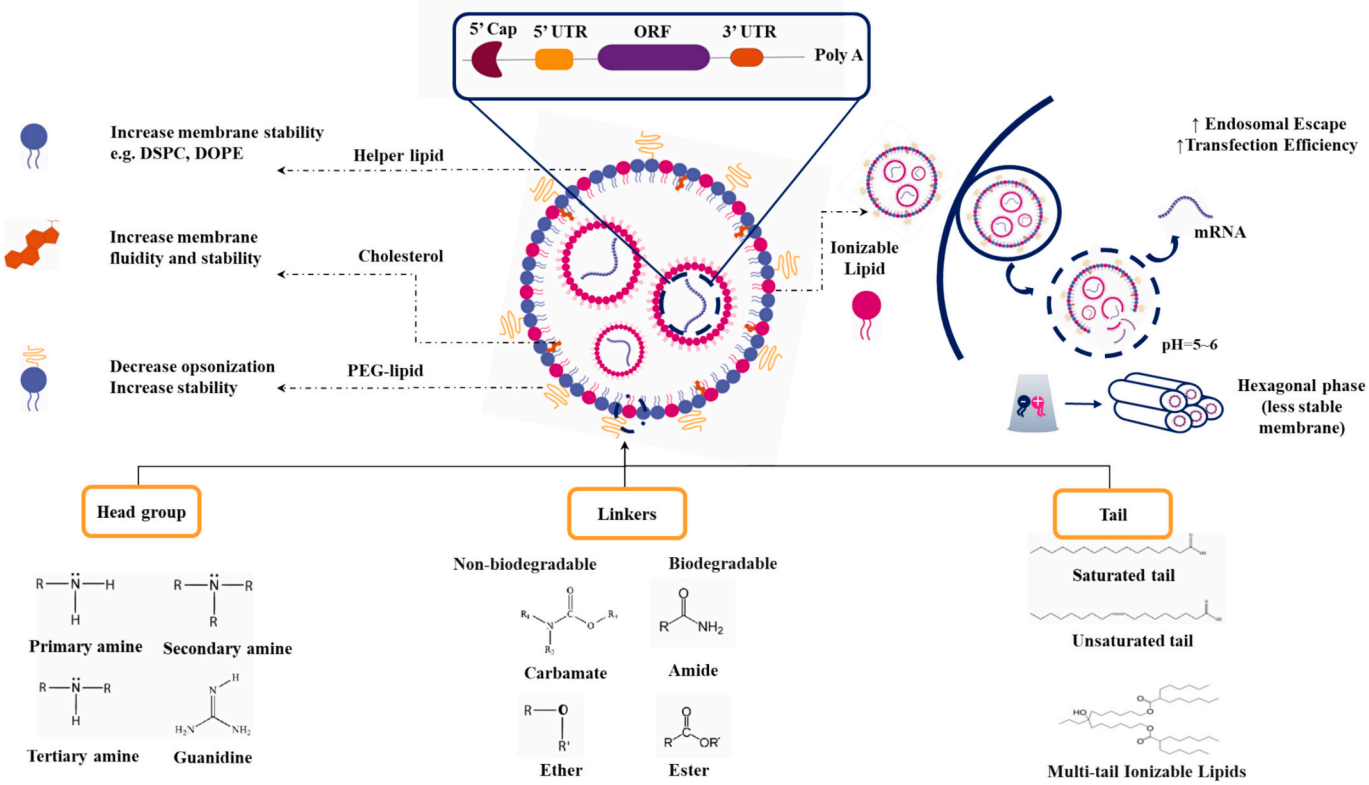


Fig. 1. Schematic illustration of lipid nanoparticles for mRNA delivery, their components and properties. The ionizable lipids can be classified based on their tails (saturated, unsaturated, multi/branched-tail), head group (amines and guanidine), and the linker (biodegradable and non- biodegradable). The interaction between ionizable lipids and endosomal membrane is essential for endosomal escape and enhancing mRNA delivery. As it shown upon cellular internalization, the acidic endosome allows protonated ionizable lipids to interact with negatively charged phospholipids, forming cone-shaped ion pairs, and inducing hexagonal structures that disrupt the endosomal membrane. This allows mRNA molecules to escape into the cytoplasm.

greater flexibility, lower immunogenicity, and cost-effectiveness [15]. Non-viral delivery methods utilize biomaterials such as lipids, polymers, peptides, biomimetic membranes, and inorganic or metal-based carriers to gene transport [16]. Various lipid-based formulations, including liposomes, lipoplexes, cationic nanoemulsions (CNEs), nanostructured lipid carriers (NLCs), and lipid nanoparticles (LNPs) have been developed for NA delivery in neurological disorders [17]. Among them, LNPs are the most advanced non-viral mRNA carriers and have emerged as promising platforms for treating NDD [18]. Recognized as a breakthrough in non-viral drug delivery [19], LNPs enable the clinical application of genetic therapies by encapsulating mRNA, protecting it from enzymatic degradation, and ensuring efficient cellular uptake [20]. Potentially, they have the potential to encapsulate mRNA encoding missing or dysfunctional proteins, anti-inflammatory cytokines, immunomodulatory agents, and neuroprotective proteins to modify disease progression.

A major challenge in mRNA delivery to the central nervous system (CNS) in NDD treatment is overcoming the blood-brain barrier (BBB) [21]. The BBB is a highly selective semipermeable barrier that restricts the passage of large molecules, including nucleic acids, making gene delivery to the brain particularly difficult [22]. The exact mechanism of LNP-mediated mRNA delivery remains debated, but it is believed to involve endocytosis, electrostatic interactions, and fusion with the cell membrane through inverted non-bilayer lipid phases [23].

To overcome these challenges, several strategies can enhance mRNA delivery, effectiveness, and brain-targeting specificity. We explored various strategies including functionalizing LNPs with targeting ligand such as monoclonal antibodies, nanobodies [24,25], peptides [26], proteins [27], and small molecules [28] that specifically bind to receptors on the BBB. Additionally, we discussed techniques like focused ultrasound combined with microbubbles [29–31], as well as the use of mannitol [32,33] to temporarily disrupt the BBB and improve drug penetration. Modifying lipid components, incorporating multimeric mRNAs, and advanced RNA modalities (circular RNAs and self-amplifying RNAs) [34,35], may further enhance gene expression.

This review explores the potential of LNPs for mRNA-based therapies in treating NDD. It examines key challenges, opportunities, and strategies to enhance delivery, targeting specificity, effectiveness, and safety. Additionally, we discussed that advancements in artificial intelligence (AI)-guided mRNA/LNP design, targeted delivery, gene editing, and CAR-T cell therapy could develop treatments for NDD and enhance clinical applications of LNP-mediated mRNA therapies.

2. Lipid nanoparticles (LNPs) and their components

LNPs are widely used in NA delivery, gaining prominence in COVID-19 mRNA vaccines (Pfizer-BioNTech and Moderna) [36,37]. By mRNA encapsulation within LNPs, mRNA protection from extracellular ribonucleases, enhanced mRNA stability, as well as efficient intracellular mRNA delivery, could be attained [38,39].

LNPs typically comprise four key components, each serving a crucial role: ionizable lipids (ILs), cholesterol, helper lipids, and polyethylene glycol (PEG)-lipids. The combination of these primary components imparts distinct functional properties to LNP systems, enabling efficient encapsulation, transport, and delivery of therapeutic payloads [40,41].

LNPs possess two crucial characteristics: their ability to accommodate a substantial amount of genetic material and their efficient release of cargo within cells. This is primarily achieved through the inclusion of ILs in the formulation, which facilitates intracellular drug release [42,43]. Fig. 1 shows the main components of LNPs and their roles and properties.

2.1. Ionizable lipids (ILs)

ILs are amphiphilic molecules used to deliver various types of NAs, including plasmid DNA, mRNA, small interfering RNA (siRNA), and antisense oligonucleotides (ASOs). These lipids can form stable complexes with NAs, protecting them from degradation and facilitating their cellular uptake [44–46]. The initial stage of self-assembly is triggered by electrostatic interactions between NA and ILs. Lipoplexes incorporating cationic lipids (CLs) have traditionally been employed for NA delivery. Nevertheless, due to concerns regarding toxicity and limited in vivo effectiveness, ILs have taken the place of CLs [39]. Due to their positive charge, CLs interact with negatively charged NAs and help encapsulate them into particles, and the strong electrostatic interaction between them hinders their release and bioavailability after cellular uptake [47].

ILs enhance the encapsulation of negatively charged NAs during LNP assembly and facilitate the delivery of cargo at endosomal pH (5.5–6.5) to cytosol [48]. ILs have pH-dependent ionization properties and undergo protonation at low pH levels, resulting in a positive charge that interacts with negatively charged phospholipids. This property leads to the formation of a cone-shaped structure, inducing hexagonal formations that disrupt the endosomal membrane and promote the release of nucleic acids into the cytosol [43]. Nevertheless, ILs remain electrically neutral under physiological pH conditions, which is advantageous for mRNA delivery in vivo. The reason is that neutral lipids exhibit fewer interactions with the negatively charged membranes of blood cells, and consequently, the biocompatibility of LNPs is enhanced [49]. This pH-dependent ionization behavior of ILs has made them a promising candidate for in vivo gene therapy and vaccine delivery applications [50–52]. The general structure of ILs can be divided into three components: the head group, the linker, and the tails [39].

2.1.1. Head group

The charge density and size of the head group play a crucial role in capturing the NA, aiding in the endosomal escape, the LNP stabilization, and the cell membrane engagement [53]. Head groups in ILs usually carry a positive charge. ILs can have multiple ionizable head groups or even a single head group. In the clinical setting, ILs like ALC-0315, DLin-MC3-DMA (MC3) [54], and SM-102 [55] possess a tertiary amine head group that demonstrates pH-dependent ionization. Additionally, the head groups of SM-102 and ALC-0315 include a terminal hydroxyl group that has the potential to decrease head group hydration and enhance hydrogen-bonding interactions with NAs, ultimately leading to a higher transfection rate [39].

2.1.2. Linker

The head group is usually connected to the tails by the linker, although in some cases, the linkers may be embedded within the tails. The presence of the linker significantly affects the stability, biodegradability, cytotoxicity, and transfection efficiency of LNPs [53]. ILs may possess one or multiple linkers. These linkers can be categorized into two groups: biodegradable linkers and non-biodegradable linkers. Biodegradable linkers such as amides, thiols, and esters are generally preferred due to their ability to be rapidly cleared in the in vivo stage, allowing for multiple doses and minimizing the likelihood of adverse effects. To minimize accumulation and potential adverse effects, ILs need to be efficiently degraded into non-toxic byproducts following successful delivery of intracellular cargo, a critical requirement for mRNA therapeutics that necessitate repeated administration. One common approach to enhance the biodegradability of ILs involves incorporating ester bonds that remain stable under physiological pH conditions but can be enzymatically cleaved within tissues and cells. For instance, to address the slow degradability of the dilinoleyl tail in MC3, a biodegradable variant (L-319) was developed by substituting one of the double bonds in each tail with a primary ester that exhibited rapid clearance and enhanced tolerability, while preserving its in vivo efficacy [56]. It is worth noting that SM-102, DLin-MC3-DMA, and ALC-0315 as the ILs in clinics, all possess ester linkers [39].

2.1.3. Tail

ILs typically consist of 1–4 hydrophobic tails with 8–20 carbon atoms each [53,57]. These tails can be either saturated or unsaturated, with the level of unsaturation playing a role in influencing NA delivery by affecting different aspects of membrane destabilization. Increasing the number of cis double bonds in the tail from 0 to 2 significantly impacts the fluidity and delivery efficiency of ILs. This change leads to a higher likelihood of bilayer lipids transitioning into a non-bilayer phase, resulting in increased membrane disruption and payload release [58]. For example, MC3 consists of two linoleyl tails. MC3 known for its robust hepatic gene silencing properties, is a critical component of the first FDA-approved siRNA drug Onpattro® for treating hereditary transthyretin amyloidosis (hATTR) [59]. On the other hand, SM-102 and ALC-0315 have two branched saturated tails that are believed to give a cone-shaped structure, aiding in the destabilization of the endosomal membrane and the release of NA into the cytosol [39].

Some structural characteristics such as the length and saturation of the lipid tails and the pKa value of the head group play a role in determining the fluidity and endosomal escape properties of formulation. The tunable charge of ILs allows for the optimization of their interactions with NAs, leading to improved encapsulation efficiency and enhanced transfection efficiency [60,61]. Overall, the development of ILs has opened up new possibilities for gene therapy and vaccine delivery, offering a promising alternative to traditional CLs for in vivo applications [45].

2.2. Cholesterol

Cholesterol plays a crucial role in the structure and function of LNPs. Cholesterol is usually incorporated into LNP structures to increase their stability and enhance their ability to deliver therapeutic payloads to target cells by filling the gaps between phospholipids and stabilizing lipid bilayers. The overall structural integrity of LNPs could also be attributed to cholesterol which provides structural support to LNPs and helps prevent premature release of the therapeutic content and protects LNPs from degradation, both in the bloodstream and after their uptake by target cells [62,63]. LNPs with a high cholesterol content tend to follow a specific pathway upon interaction with Apolipoprotein E (ApoE) and subsequent processing through the low-density lipoprotein (LDL) pathway [64]. The involvement of cholesterol in endosomal processing has also been linked to its role in LNPs as cholesterol can lead to the enhancement of endosomal escape [48]. However, excessive cholesterol levels in LNP formulations can lead to the formation of soluble cholesterol crystallites within the LNP core, along with deprotonated entities, due to surpassing the stable retention threshold at the molar level [65,66]. Recent research indicates that natural sterol alternatives have the potential to replace cholesterol in LNP formulations. Cholesterol derivatives, such as phytosterols and β-sitosterol derivatives, have shown promise in enhancing the effectiveness of LNPs by possessing longer alkyl chains and improving fusogenic and endosomal escape properties [45,67].

2.3. Helper lipids

To enhance the stability and fluidity of LNPs, helper lipids are essential, which provide support for stability during storage and circulation as well as blood compatibility. The term “helper lipids” encompasses various lipids, such as phospholipids, glycerolipids, and sterols [67]. The two most frequently utilized phospholipids in LNP formulations as helper lipids are Di stearoyl phosphatidylcholine (DSPC) and 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine (DOPE). Phospholipids such as DOPE or DSPC have the ability to enhance the permeability of the cell membrane, thereby aiding in the uptake of cells and escape from endosomes [15]. DSPC is a saturated phosphatidylcholine (PC) lipid

with a high transition temperature (Tm). This contributes to the formulation's prolonged circulation time, overall stability, and induction of cell-mediated immunity [68]. Conversely, DOPE, an unsaturated lipid, exhibits a truncated cone-shaped structure due to the presence of double bonds in its acyl chains and a smaller head group compared to the tertiary (trimethylated) amine group in PC. These structural characteristics of DOPE facilitate the formation of the non-bilayer hexagonal (HII) phase due to its cone-shape geometry, thereby enhancing the intracellular delivery of NAs (by endosomal release) [20,44,67,69].

Non-ionic surfactants, such as monooleoylglycerol (MOG) [70], diolein [71], and SPAN80 [72], are also considered helper components and are capable of enhancing non-bilayer hexagonal II or cubic phase transition. Additionally, for cationic LNPs, triolein has been considered as a helper component [67]. Typically, these lipids are non-cationic, although the term has also been extended to include surfactants and PEG-lipids. CLs such as 1,2-dioleoyl-3-trimethylammonium-propane (DOTAP) or N-[1-(2,3-dioleoyloxy)propyl]-*N*,*N*,*N*-trimethylammonium chloride (DOTMA), can also be included as helper lipids in LNPs. CLs can enhance the cellular uptake and endosomal escape of LNPs. These CLs interact with the negatively charged cell membranes, facilitating the internalization of LNPs and the release of their cargo into the cytoplasm [51,73]. It is worth noting that the molar ratio for helper lipids in LNPs is lower than the molar ratio for ILs [44]. Overall, including helper lipids in LNPs is crucial for their stability, fluidity, cellular delivery and allows for optimizing LNP formulations for specific applications [74].

2.4. Polyethylene glycol (PEG)-lipids

The inclusion of PEGylated lipid such as 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethyleneglycol)-2000] (DSPE-PEG_{2000}) in LNPs is common, although present at a minor molar percentage in LNPs. PEG-lipids are often used to improve the stability and circulation time of LNPs. DSPE-PEG_{2000} provides a hydrophilic coating on the surface of LNPs, preventing opsonization and clearance by the immune system. Incorporation of PEGylating lipids also helps to reduce the aggregation and fusion of LNPs [67,74], improve tolerability by reducing immune recognition and clearance [75], and further enhancing their stability [67,74].

PEGylation of nanoparticles decreases access to the LNP surface, reduced opsonization and uptake by the reticuloendothelial system (RES), thereby prolonging the circulation of LNPs. In addition, it facilitates ligand attachment to LNPs. Factors such as PEG molecule size, architecture, carbon tail length, and PEG-lipid molar ratio affect the effectiveness of LNP delivery systems [76,77]. However, PEGylation of nanoparticles can hinder the cellular uptake of nanoparticles and impede their escape from endosomes, thereby reducing overall efficacy by creating a steric barrier that obstructs efficient particle binding and release from endosomes [78,79].

ALC-0159 is a PEGylated lipid (N,N-dimyristylamide of 2-hydroxyacetic acid, conjugated to a PEG chain), which considered as a non-ionic surfactant and has been deployed in the Pfizer-BioNTech SARS-CoV-2 mRNA vaccine [80].

Recent patents have also examined the potential of polyglycerol [81,82], poly(*N*-methyl-N-vinylacetamide) (PNMVA) [83] and polysarcosine as an alternative to PEG, offering biocompatibility, adjustable properties, improved renal clearance, and biodegradability [84]. In comparison, polyglycerol does not increase blood viscosity, enhances circulation duration, and avoids accelerated blood clearance [81,85]. Despite the advantageous properties and minimal immunogenic reactions of alternative polymers, none of them have been found to be more effective than PEG in enhancing the pharmacokinetic performance of LNPs [86,87].

3. mRNA-LNP delivery

LNPs containing mRNA, produced through rapid mixing, demonstrate a stable nanostructure that can encapsulate mRNA within the core via electrostatic interactions with the lipids [88]. This unique structure shields the mRNA from degradation by nucleases and enhancing stability of LNPs in biological fluids [49,89]. LNP-mRNA formulations must overcome several obstacles (extracellular and intracellular) to operate within a living organism. It is important for LNP-mRNA formulations to evade the mononuclear phagocyte system (MPS) and renal clearance, target and enter specific cells, and escape endosomes to reach the cytoplasm for translation [49,90].

Once LNPs reach the target cells, they can be internalized through various mechanisms, including macropinocytosis, clathrin-mediated endocytosis, and caveolae-mediated endocytosis, depending on nanoparticle properties and the cell type. Following internalization, the lipid nanoparticles are typically trapped within endosomal compartments [91]. Ionizable lipids are anticipated to promote an inverted cone-shaped structure, enhancing endosome-escape capability. As mentioned, this pH-dependent ionization behavior of ILs has made them a promising candidate for in vivo gene therapy of various diseases and vaccine delivery applications.

3.1. mRNA structure

mRNA molecule facilitates the efficient translation of the DNA genetic code into proteins by the ribosomes in the cytoplasm, and can be engineered to produce any therapeutic protein, offering a strategy to address previously considered "undruggable" conditions. Non-replicating mRNA comprises of an open reading frame (ORF) bordered by a 5′-prime region and a 3′-prime untranslated region (UTR), a 5′ cap structure, and a 3′ poly (A) tails [92].

The ORF is a crucial segment of mRNA that is responsible for encoding proteins. It consists of a series of uninterrupted codons between the start and stop codons. The ORF plays a significant role in determining the amino acid sequence of the protein that is ultimately produced, directly impacting mRNA stability and translation efficiency [93] while UTR sequences are responsible for maintaining the stability of mRNA, facilitating efficient and accurate translation, aiding in the recognition of mRNA by ribosomes, determining subcellular positioning, and facilitating communication between mRNA and its associated proteins [14]. Meanwhile, poly (A) tails are essential 3′ modifications that interact with proteins to regulate mRNA stability and translation. Their length affects translation efficiency and plasmid stability [94,95]. Recent research has shown that the 5′ cap and 3′ poly(A) tail of mRNA cooperate to control mRNA translation and stability [96]. Poly(A) tails is crucial in enhancing mRNA stability and promoting protein synthesis [14]. The 5′ cap is essential for stability, minimizing immunogenicity, and protecting mRNA from cellular exonucleases [14].

3.2. The LNP-mediated mRNA delivery for neurodegenerative-diseases treatment

mRNA-based therapies have shown promising potential for tackling NDD, offering a new avenue for effective treatment. NDD can result from genetic mutations leading to protein deficiencies. Therefore, LNPs can encapsulate mRNA encoding the missing or dysfunctional protein, facilitating its translation within neuronal cells. This approach has shown promise in preclinical models and could potentially restore normal function. Moreover, some NDD have an inflammatory component. LNPs can be used to deliver mRNA that encodes anti-inflammatory cytokines, immunomodulatory agents, and neuroprotective proteins to alter the disease progression. LNP-encapsulated mRNA could be used to create vaccines targeting pathological proteins, such as amyloid-beta in Alzheimer's disease, to stimulate an immune response. In addition, LNPs can be engineered to target specific cell types within the brain, such as microglia or neurons, by modifying their surface properties or incorporating targeting ligand. This specificity can enhance therapeutic efficacy while minimizing off-target effects. The successful delivery of

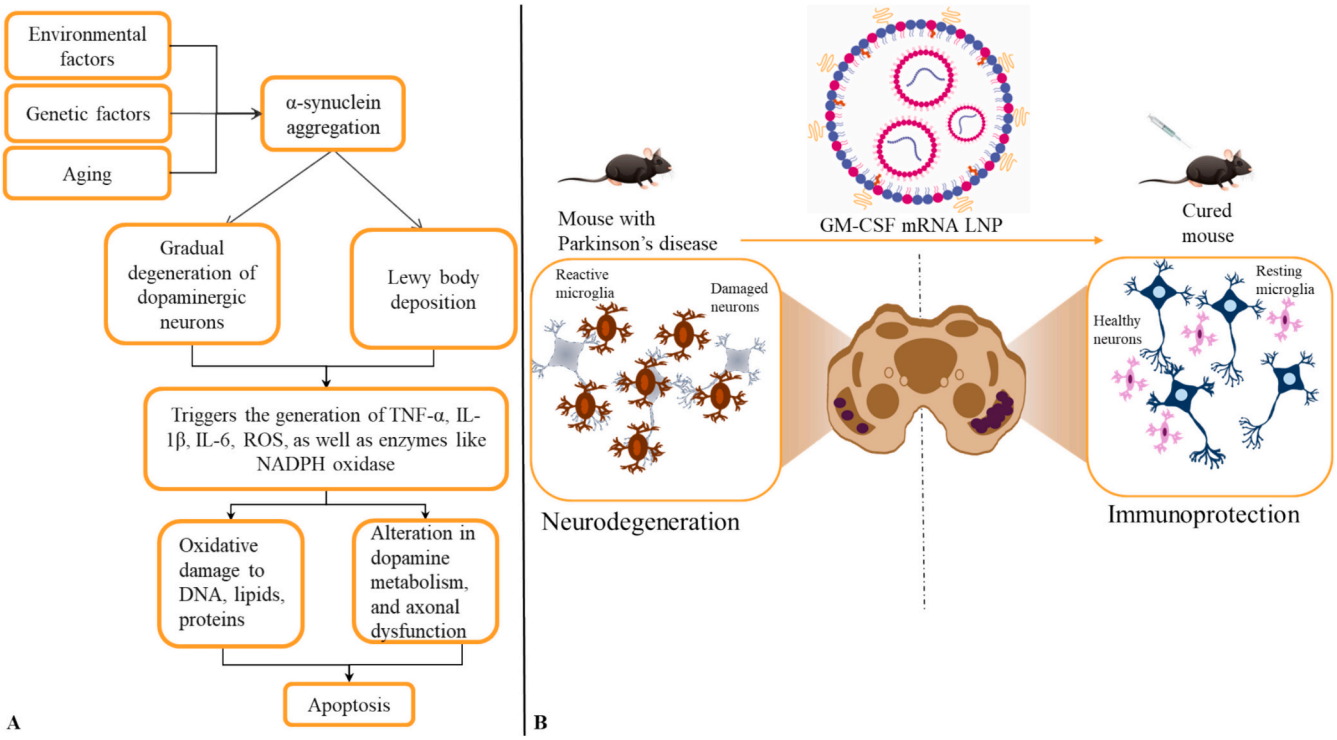


Fig. 2. Schematic representation of (A) Parkinson's disease pathophysiology. Parkinson's is characterized by the accumulation of α-synuclein, forming Lewy bodies. (B) Its proposed treatment: mRNA for neurotrophic factors can promote the survival of dopaminergic neurons. GM-CSF mRNA administration in PD-like mouse models resulted in a dose-dependent increase in GM-CSF levels, Treg populations, immunosuppressive mRNA expression, and CD_4^+ T cells. These changes corresponded with the preservation of dopaminergic neurons in the substantia nigra and striatum.

mRNA-based therapeutics using LNP-based vectors has the potential to change the treatment of neurological disorders. Here, we have discussed the pathophysiological mechanisms of neurological disorders that have been treated by different mRNA-LNP formulations.

3.3. *Parkinson's disease (PD)*

Parkinson's disease (PD), described by James Parkinson in 1817, is a neurodegenerative disorder that affects movement, balance, and coordination in elderly [97]. The hallmark symptoms of PD include tremors, rigidity, and bradykinesia (slowness of movement). It is the second most common NDD associated with aging that imposes a heavy physical, psychological and financial burden on patients [98,99].

3.3.1. *Manifestations and pathophysiology of Parkinson's disease*

A combination of environmental and genetic factors can cause PD. Exposure to toxic chemicals, pesticides, and head injuries can increase the risk of developing PD [100,101]. PD is considered among the neurodegenerative disorders that affect the neurons of the human brain, resulting in the deterioration of brain function [102]. One possible cause of PD could be the gradual degeneration of dopaminergic neurons located in the substantia nigra, along with the emergence of Lewy bodies within neuron cytoplasm, which are cytoplasmic inclusions containing aggregated α-synuclein [103]. Another possible cause of PD could be α-synuclein, a soluble protein consisting of 140 amino acids and has a predicted molecular weight of approximately 14,000 Da [104], which can be divided into three distinct regions: an amino-terminal lipid-binding region, a central hydrophobic non-amyloid-β region known as the non-amyloid-β component domain, and a disordered carboxy-terminal region [105]. Environmental and genetic factors as well as aging could lead to the α-synuclein aggregation. The aggregation process contributes to demise of neuronal cells and the emergence of persistently activated glial cells, namely microglia and astrocytes. The amount of α-synuclein present is thought to play a crucial role in the formation and accumulation of these fibrils [106]. This phenomenon subsequently triggers the generation of tumor necrosis factor alpha (TNF-α), interleukin-1 beta (IL-1β), interleukin-6 (IL-6), reactive oxygen species (ROS), as well as enzymes like nicotinamide adenine dinucleotide phosphate (NADPH) oxidase [107]. Furthermore, non-neuronal cells play a significant role in the advancement of PD pathology [103].

It has been found that there is a notable rise in α-synuclein-specific T cells among individuals diagnosed with PD [108,109], which linked to the specific type of the human leukocyte antigen (HLA) system, indicating a potential autoimmune role of T cells in the development of PD [109].

Neuroinflammation plays a significant role in the development and progression of PD. In various animal models of PD, activated microglia, CD_4^+ T cells, and CD_8^+ T cells have been observed in the substantia nigra pars compacta (SNpc) with a more predominant role of CD_4^+ T cells in PD progression compared to CD_8^+ T cells. This is supported by studies showing that mice lacking CD_4^+ T cells experience alleviated neuronal loss, whereas mice lacking CD_8^+ T cells do not [110].

More recent research has focused on the dysregulation of specific subsets of CD_4^+ T cells (T helper 1 (Th1), T helper 2 (Th2), T helper 17 (Th17), regulatory T cell (Treg)) in the peripheral blood of PD patients [111]. PD patients have been found to have increased percentages of Th17 and Th1 cells (inflammatory T cells), as well as decreased percentages of Treg and Th2 cells (anti-inflammatory T cells) in their peripheral blood with impaired function of Tregs. Th17 cells have been shown to exacerbate the loss of dopaminergic neurons and axonal dysfunction in PD [110].

Another critical indicator of PD development is the overproduction and inability to detoxify ROS and reactive nitrogen species (RNS) effectively, which both lead to oxidative and nitrative stress in PD which promote apoptosis and the degeneration of dopaminergic neurons. The disruption of these key components in the PD substantia nigra is

followed by dysregulation of dopamine, iron, and calcium metabolism, an increase in neuroinflammatory cells, aging and mitochondrial dysfunction [112–114]. Genetic mutations, especially in the proteins associated with PD (inherited or environmentally) can also lead to the development of PD. Targeting the underlying genetic causes of PD could slow down the degeneration of dopaminergic neurons [115].

While the cause of the disease is known, symptoms of the disease only manifest after extensive neuron loss, highlighting the importance of early cause identification [112,116]. Fig. 2A shows the main pathophysiology suggested for Parkinson's disease.

3.3.2. PD therapies

Despite significant advancements made in recent decades, PD remains an incurable ND and the second largest in terms of prevalence. In all patients, treatment of PD focuses on improving both motor and non-motor signs and symptoms through symptomatic means [117], primarily using drugs such as Levodopa formulations, dopamine agonists, and monoamine oxidase-B (MAO-B) inhibitors, which are often supplemented with deep brain stimulation or nerve nucleus destruction [118].

In the past, anticholinergic agents such as trihexyphenidyl were considered beneficial for young individuals with significant tremors. However, current research recommends early Levodopa treatment for PD despite its potential for dyskinesia especially at higher dosages [119]. On the other hand, MAO-B inhibitors and dopamine agonists offer less substantial relief from symptoms but are linked to a reduced risk of dyskinesia. Dopamine agonists, in particular, are connected to a higher overall incidence of adverse effects. Finally, the majority of PD-suffering patients rely on medications from various classes to achieve complementary advantages while minimizing the need for high doses and dose-related adverse reactions [120]. Therefore, there is a significant demand for platforms that can enhance bioavailability, while decreasing dosing frequency for more effectiveness of treatments [117].

It is noteworthy that there has been much developments in NA-based therapies for PD. For instance, genetic regulation with particular focus on downregulation of α-synuclein gene using microRNA, siRNA, DNA aptamers, and vector-assisted delivery systems [74].

The development of nanomedicine for therapeutic delivery to the brain could provide an alternative approach to the current drug treatments. Nanoparticles offer a range of benefits in neuroprotection and neurorepair due to their adjustable physical and chemical properties, as well as their ease of modification and functionalization [117,121].

Restoring the numbers and functionality of regulatory T cells (Tregs) presents a novel therapeutic approach for neurodegenerative conditions, providing a potent anti-inflammatory response that helps protect neurons. In this regard, a LNP was created to carry mRNA encoding granulocyte-macrophage colony-stimulating factor (GM-CSF mRNA), which was used to stimulate Tregs peripherally in preclinical models of PD and considered a potential therapy for PD. GM-CSF mRNA LNP administration in mice and rats overexpressing α-synuclein caused dose-dependent enhancement of plasma GM-CSF levels as well as peripheral CD_4^+,CD_{25}^+, $FoxP_3^+$, Treg populations, which resulted to nigrostriatal neuroprotection, and diminished reactive microgliosis [122]. Due to GM-CSF short plasma half-life, this platform was amenable to extending GM-CSF half-life and thus its bioavailability [123].

When GM-CSF mRNA was administered to mice and rats with PD-like symptoms, there was a dose-dependent increase in plasma GM-CSF levels and peripheral CD_4^+, CD_{25}^+, $FoxP_3^+$, Treg populations. This increase in Tregs was associated with the protection of the nigrostriatal pathway, upregulation of immunosuppression-related mRNAs, detection of a CD_4^+ T cell population induced by treatment, and reduced reactive microglial response. Consequently, this led to the safeguarding of dopaminergic neurons in both the substantia nigra and striatum regions [124].

Nevertheless, despite the proven effectiveness of GM-CSF mRNA delivery, the cytokine's limited half-life, poor bioavailability, and adverse reactions at the injection site posed challenges for wider application. To address these constraints, further researches developed mRNA-LNP formulations containing a fusion protein of extended half-life albumin-GM-CSF for both mouse (Msa-GM-CSF) and rat (Rsa-GM-CSF) models. In fact, albumin conjugation provides a longer half-life for GM-CSF and mRNA LNP also encodes an extended half-file GM-CSF fusion protein. These formulations were assessed for their immunomodulatory and neuroprotective effects using preclinical models of PD involving acute 1-methyl-4-phenyl-1, 2,3,6-tetrahydropyridine (MPTP) and chronic human wild-type α-synuclein overexpression. A single administration of the extended half-life mouse and rat mRNA LNPs resulted in detectable GM-CSF plasma cytokine levels for up to four days. The increased frequency and function of Treg cells were linked to a quiescent microglial state, protection of nigrostriatal neurons, and restoration of brain tissue immune balance. These outcomes surpassed the known efficacy of daily recombinant wild-type GM-CSF with a half-life of six hours. Neuropathological transcriptional profiles in the affected brain region revealed upregulation of neuroprotective CREB (cAMP-response element binding protein), synaptogenesis signaling, as well as neurovascular coupling pathways. These results underscore the enhancement in therapeutic efficacy associated with the mRNA-encoded albumin GM-CSF fusion protein modification, which was correlated with improved bioavailability of the medication. Albumin-GM-CSF fusion protein mRNA LNP could be a highly effective therapeutic when administered as a single dose once a week in both a chronic and acute animal model of PD. Notably, mRNA LNP encoding the extended half-life albumin-GM-CSF fusion protein could set a standard for immune-based therapies in PD, particularly in patients with tremors and gait abnormalities [125]. Fig. 2B summarizes the outcome of GM-CSF mRNA-based LNP for PD treatment.

3.4. Multiple sclerosis (MS)

Multiple sclerosis (MS) is an autoimmune and chronic inflammatory disease that is associated with the abnormal function of the immune system with prominent CNS hallmarks [126]. A variety of factors influences the development of MS such as age (average onset at 20–40 years) and gender [127] with the higher prevalence among women. Epidemiological studies, genetic and environmental factors also play significant roles in the prevalence of MS [127,128].

3.4.1. Manifestations and pathophysiology of MS

Numerous studies indicate that autoimmune mechanisms involving cellular immunity (Th1, Th17) and humoral immunity (B cells) significantly contribute to the progression of MS. B-cells affect the pathogenesis of MS via T-cell regulation and production as well as APC activity. MS is mainly characterized by various signs such as demyelination primarily in the spinal cord, optic nerve, or brain stem [126]. Like other cell types, immune cells require a constant supply of amino acids (AAs) to sustain their metabolic functions and overall viability. Upon activation by inflammatory or antigenic stimuli, the demand for AAs in these cells escalates sharply. While a lack of AAs can impair the immune response under certain circumstances, immune cells also possess AA-sensitive pathways that can trigger substantial metabolic changes in response to fluctuations in AA availability. Additionally, to the genetic history of MS and some viral infections that may induce MS, diminished AA catabolism can also be a predisposing factor for MS. The basal catabolism of AAs may play a role in maintaining immune homeostasis and preventing autoimmune disorders, whereas heightened AA catabolic activity could promote immune suppression. Furthermore, there is growing evidence that certain downstream metabolites derived from AAs serve as significant biological regulators of autoimmune responses. Among the AAs that influence immune activity, L-Tryptophan (Trp) and L-Arginine (Arg) are particularly noteworthy. Tryptophan is metabolized by the enzymes 2,3-dioxygenase (TDO) and indoleamine 2,3-dioxygenase (IDO) 1 and IDO2, while Arg is catabolized by inducible nitric oxide

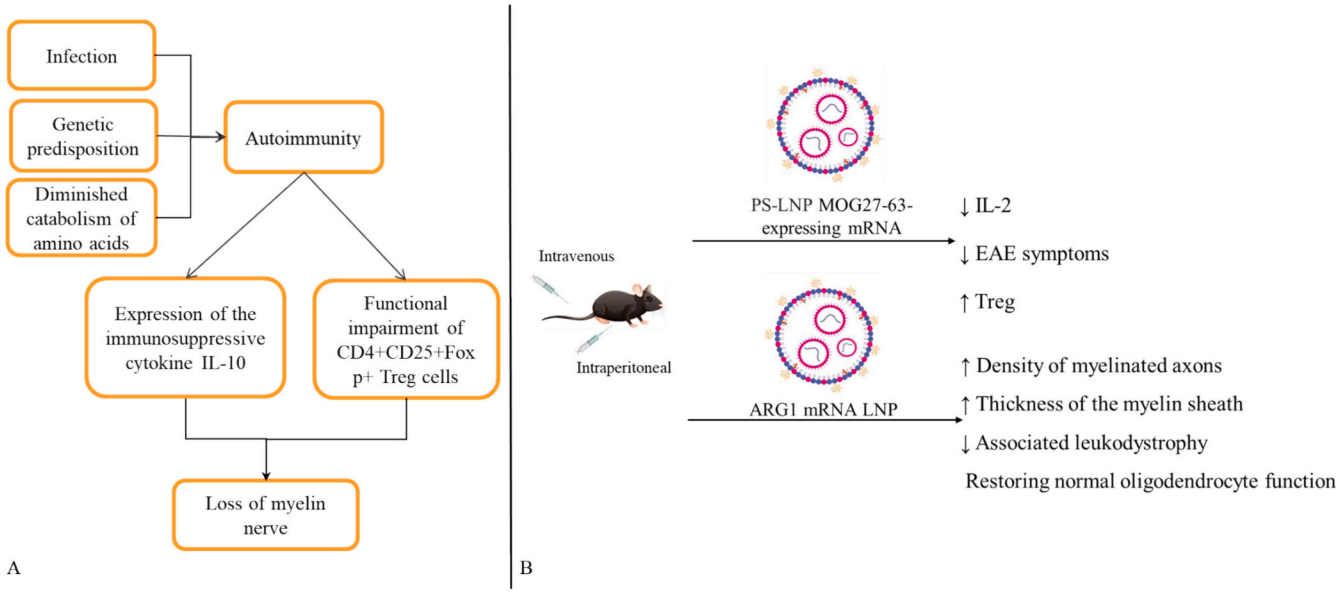


Fig. 3. Schematic representation of (A) Multiple sclerosis pathophysiology, (B) therapeutic effect of mRNA-LNPs against experimental autoimmune encephalomyelitis (EAE model) 1. The PS-LNPMOG demonstrated a more significant reduction in the $CD4^+/CD8^+$ ratio compared to the PS-LNP Luc and inhibited T-cell infiltration into the mouse brain 2. ARG mRNA-based LNP treatments on the other hand increase the density of myelinated axons and decrease leukodystrophy.

synthase (iNOS) and two isoforms of arginase (ARG1, ARG2) [129]. Two arginases, ARG1 and ARG2, encoded by distinct genes, catalyze the breakdown of Arginine in myeloid cells such as macrophages, dendritic cells, and activated T cells and both are contributed to the establishment of an anti-inflammatory microenvironment that supports tissue repair Research indicates that the catabolism of AA is diminished in MS-suffering patients, and this reduction has significant functional implications, leading to an increase in pro-inflammatory cytokines and a decrease in regulatory T cell (Treg) populations [129].

Additionally, the functional impairment of natural CD_4^+, CD_{25}^+, $Foxp_3^+$, Treg cells in MS patients was also demonstrated [130]. The expression of the immunosuppressive cytokine IL-10 is reduced in MS patients, consistent with previous studies reporting a decrease in IL-10-secreting peripheral blood mononuclear cells (PBMC). Interestingly, there is a decrease in the expression of IL-10 but an increase in the number of PBMC expressing IL-10 transcripts prior to clinical relapse in MS [131]. Fig. 3A shows the main pathophysiology suggested for multiple sclerosis.

3.4.2. MS therapy

Disease-modifying medications like dimethyl fumarate, interferon-β, natalizumab, and fingolimod are commonly used in the treatment of MS [132]. The primary objectives of MS management are to minimize lesion activity and prevent secondary progression. Current treatment strategies focus on managing acute episodes, alleviating symptoms, and reducing the biological activity of the disease. Traditionally, MS has been classified as an organ-specific T-cell mediated autoimmune disease. However, the success of B-cell targeted therapies challenges this standard T-cell autoimmune dogma. Treatment aims to regulate autoreactive T cells without causing widespread immune suppression [126,133,134].

A critical obstacle lies in effectively delivering specific antigens in a manner that does not trigger inflammation. LNP containing ARG1 mRNA has demonstrated promise in this regard, significantly improving myelin density, restoring normal oligodendrocyte function, and prevents leukodystrophy in those with arginase deficiency (studied in Arg1 knockout mice). This study is further suggesting an increase in myelinated axons and thicker myelin sheaths as the pathway for ammonia detoxification in terrestrial mammals, is mediated by ARG1. This type of LNP also elevated the level of arginine, guanidine, and ammonia in the plasma of the treated mice [135,136].

Previous research has shown that LNPs containing phosphatidylserine (PS) can effectively deliver $MOG_{27–63}$-expressing mRNA (as the self-antigen mRNA) to the spleen of the experimental autoimmune encephalomyelitis (EAE) mice, resulting in the expression of desired proteins [137]. By the virtue of PS presence in this type of LNP, it showed its ability to evade immune system activation by suppressing immune responses against apoptotic cells. This LNP type also caused recovery of the CD_4^+/CD_8^+ cell balance, thus ameliorating various aspects of EAE such as clinical symptoms and autoimmunity induction related cells [137]. When PS is recognized on apoptotic cells, it suppresses immune activation against these cells by releasing anti-inflammatory cytokines like IL-10 and transforming growth factor (TGF)-β. Another study aimed to investigate whether mRNA delivery of autoantigens to the spleen using PS-LNPs could induce the expression of $MOG_{35–55}$ antigens with minimal immune stimulation. Furthermore, the study aimed to determine if this approach could be utilized to EAE as the prevalent animal model of MS, by inducing immune tolerance [138]. On the other hand, the specific delivery of autoantigens to $CD11c^+$ antigen-presenting cells (APCs) residing in lymphoid tissues leverages a highly effective natural mechanism to induce and uphold peripheral tolerance. The exposure of autoantigens in a non-inflammatory setting triggers the expansion of antigen-specific CTLA-4^+, $ICOS^+$, IL-10^+, and $Foxp_3^+$ effector Tregs, which not only suppress auto-reactive effector T cells but also induce bystander immunosuppression, thus facilitating disease control even in a complex, polyclonal autoimmune model [139]. A LNP system for delivery of mRNA into B cells (for specific targeting of the spleen) was designed, which encapsulates mRNA, navigates to the spleen, and transfects B lymphocytes, inducing protein expression. The LNP induced more than 85 % of total protein production in the spleen, despite transiently appearing in the liver and other organs in vivo. This platform was designed based on the fact that, B cell dysregulation is implicated in autoimmune disorders (especially MS) and cancers (non-Hodgkin's lymphomas). Thus, this type of mRNA-LNP could be candidate for treatment of MS based on mRNA-LNPs [140,141]. The effect of ARG_1mRNA-LNP and PS-LNP MOG mRNA-based LNP for MS treatment in mice model is summarized in the Fig. 3B.

3.5. Spinal Cord injury (SCI)

Spinal trauma encompasses injuries to the spinal cord or the

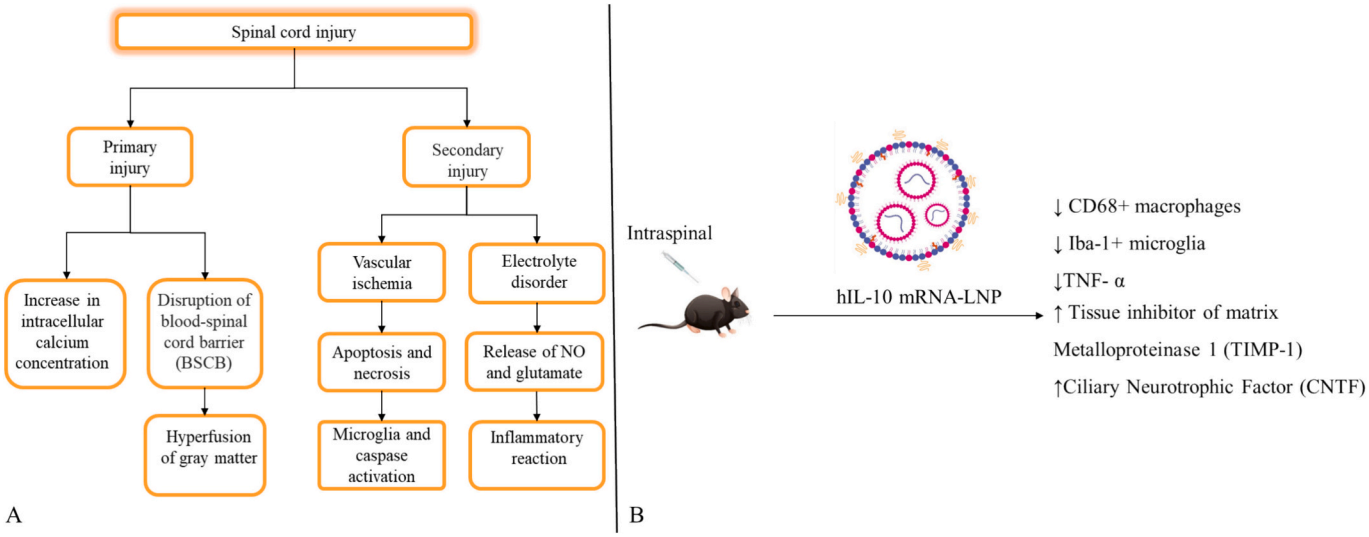


Fig. 4. Schematic representation of (A) spinal cord injury pathophysiology, (B) the effect of mRAN-based LNP treatments on spinal cord injury: mRNA-LNPs encoding a combination of therapeutic proteins (neurotrophic factors like brain-derived neurotrophic factor (BDNF), or fibroblast growth factor (FGF)) can promote both structural and functional recovery after spinal cord contusion and led to a significantly smaller lesion area.

surrounding bones and tissues, leading to significant disability on a global scale, particularly following severe traumatic events [142].

3.5.1. *Manifestation and pathophysiology of SCI*

SCI is characterized by a complex interplay of cellular and molecular processes that hinder regeneration and cellular viability. The disruption of interactions among neuronal cell types, including neurons, astrocytes, microglia, oligodendrocyte progenitor cells (OPCs), and oligodendrocytes, underlies the clinical manifestations and recovery challenges associated with SCI. SCI can be divided into primary and secondary injury phases. The primary phase involves immediate physical trauma, causing hemorrhage, edema, axotomy, and cell death in various cell types through necrotic processes [143]. This damage leads to increased pressure at the injury site, exacerbating ischemia and negatively affecting local neurons and glial cells. Disruption of the Na^+/K^+ pump function contributes to cellular swelling and further cell death [143,144]. During the secondary phase, elevated expression of inhibitory proteins and insufficient trophic support create an unfavorable environment for cell survival and regeneration. This phase involves various detrimental processes, including apoptosis, demyelination, inflammation, and astrogliosis, which contribute to the formation of a glial scar. Lasting for months, this phase results in additional cellular death, damage, and a non-functional spinal cord [143].

The release of pro-inflammatory cytokines by activated macrophages and neutrophils prompts reactive astrocytes to increase the secretion of inhibitory molecules like chondroitin sulfate proteoglycans at the injury site [145,146]. Inflammation in the injured spinal cord causes swelling, leading to compression within the vertebral column and subsequent secondary damage. Furthermore, reactive astrocytes eventually form a glial scar at the lesion site, acting as a mechanical and chemical barrier to axonal regrowth [147]. Inflammation is also crucial for regeneration, with macrophages playing a key role. These macrophages, derived from resident microglia and hematogenous sources, exhibit plasticity and can switch between pro-inflammatory M1 phenotype (or classically activated) and pro-regenerative M2 phenotypes (or alternatively activated) based on the microenvironment [148,149]. The pro-inflammatory phenotype is induced by prototypical Th1 cytokines such as interferon-γ (IFN-γ), which trigger the production of pro-inflammatory cytokines, ROS, and nitric oxide (NO), and contribute to inflammation and tissue damage. On the other hand, the pro-regenerative phenotype is associated with Th2 cytokines; Invading macrophages release cytokines (IL-10, IL-4, IL-6) that can influence the expression of pro-inflammatory molecules such as IL-1, IL-2, IL-6, TNF-α, IFN-γ, matrix metalloproteinase-9, nitric oxide synthase, myeloperoxidase, and ROS is reduced by IL-10. IL-10 also promotes the survival of neurons by interacting with the IL-10 receptor. Immediate systemic administration of IL-10 following SCI leads to enhanced tissue preservation, functional improvement, and neuroprotection macrophages infiltrate the lesion area and play dual roles in injury and repair processes [150]. These macrophages exhibit a spectrum of activation states, ranging from pro-inflammatory to pro-regenerative. Proregenerative macrophages exhibit heightened phagocytosis capabilities and can facilitate tissue regeneration. The balance between proinflammatory and proregenerative macrophages can either enhance or hinder axonal growth towards the lesion site [147,151]. Recovery from SCI is hindered by the physiological changes and neuronal loss that occur after the initial injury. In the days and weeks following SCI, the damaged tissue experiences inflammation and apoptosis, leading to the loss of myelin and the formation of a glial scar that inhibits the growth of new axons. This secondary injury significantly impairs the recovery of motor function [152]. SCI pathophysiology is summarized in the Fig. 4A.

3.5.2. *Spinal cord injury therapy*

The damage caused by vascular rupture after SCI is exacerbated by neurogenic shock, which is managed using vasopressin and cardiac stimulants such as dopamine, norepinephrine, atropine, and midodrine. Additionally, the use of the sulfonylurea receptor 1 channel antagonist glibenclamide suggests prevention of further damage to the spinal cord microvasculature, thereby preventing progressive necrosis and improving outcomes. Methylprednisolone is commonly used to treat secondary injury after SCI, but it has limited effectiveness and adverse effects [153,154]. Tirilazad, a non-glucocorticoid steroid, has been explored as an alternative but has shown less effectiveness [155]. Similarly, Minocycline, an anti-inflammatory compound, shows promise in clinical use [156]. Moreover, acute baclofen treatment has been found to improve sensory and motor function in SCI patients. Despite years of scientific research, no medication has proven effective in phase III clinical trials, underscoring the importance of prioritizing preclinical studies on pharmacological interventions aimed at enhancing neurological function in SCI [153]. Given that the initial injury is irreversible, preventing further damage is crucial to maintain structural integrity and minimize functional impairment. The administration of IL-10 protein has shown promising results in the treatment and potential cure of SCI. It is widely recognized that IL-10 possesses the capability to inhibit the

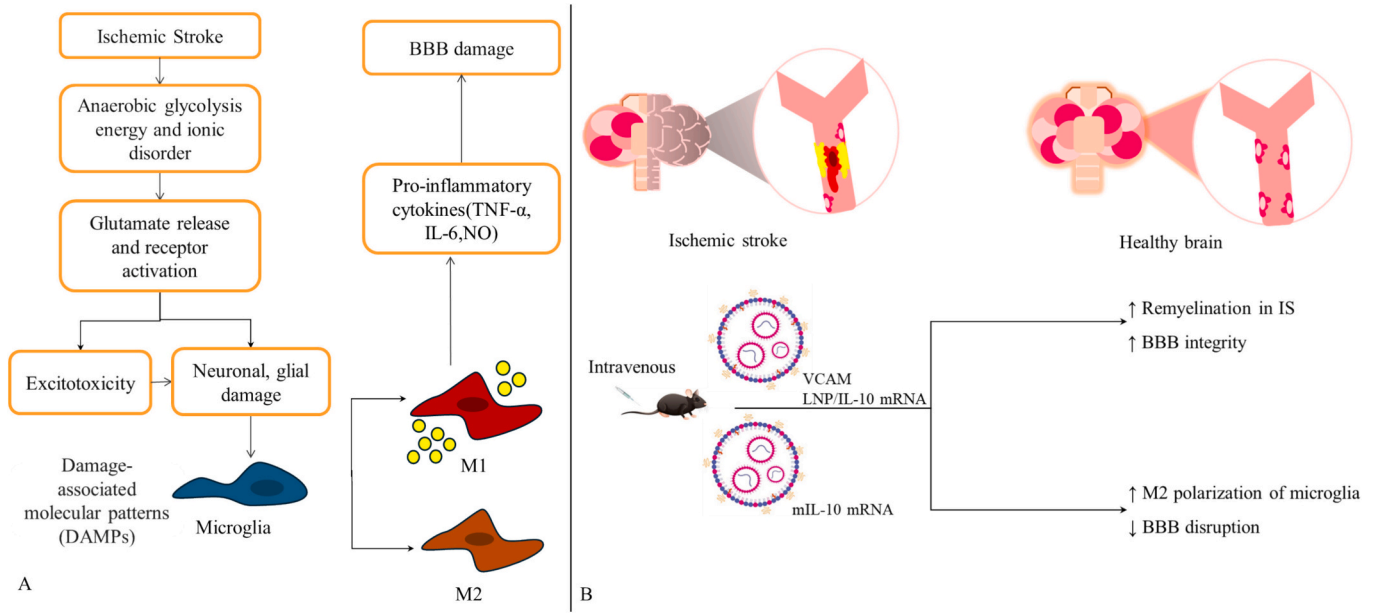


Fig. 5. Schematic representation of (A) ischemic stroke pathophysiology: One immediate response to stroke is the release of excitatory neurotransmitters like glutamate, which in turn can cause neuronal damage. During the initial phase of stroke, pro-inflammatory microglia M1 lead to an increase in inflammatory substances like cytokines, chemokines, and ROS, while M2 phenotype through specific signaling pathways, thereby resolving neuroinflammation, protecting the BBB, and restoring neurological function post-stroke (B) The proposed treatment: vascular cellular adhesion molecules (VCAM)-targeted LNPs and IL-10 mRNA-LNP exhibited selective expression in the ischemic brain region, and promote M2 polarization of microglia. mRNA encoding neuroprotective proteins can promote neuronal survival, reduce cell death, and enhance recovery following stroke.

expression of inflammatory cytokines and the activation of inflammatory macrophages in the damaged spinal cord [10,151]. IL-10 hampers cellular immunity and hinders the production and discharge of proinflammatory agents like TNF-α, IL-1β, IL-6, IL-8, and IL-12 [10,151,157].

In this regard, it has been demonstrated that a combination of recombinant human interleukin-10 (hIL-10), recombinant human interleukin-6 (hIL-6), macrophage inflammatory protein 1-alpha (hMIP-1-alpha), and glial cell line-derived neurotrophic factor (hGDNF) administered through an osmotic pump can promote neuroprotection and enhance functional restoration after spinal cord contusion injury spinal cord tissue in rats [158]. Following that study, the intraspinal delivery of anti-inflammatory proteins as mRNA-LNP was applied one-week post-contusion injury. It has been shown that mRNA-LNP administration could modulate the injured spinal cord microenvironment and could result in the modulating IL-10, migration of macrophages and lymphocytes and promote neuroprotection [10]. Functional assessments revealed that the therapeutic proteins produced by the mRNAs improved coordinated movement in the treated group compared to the control group. Moreover, the administration of mRNA-LNPs led to a significantly smaller lesion area at the injury epicenter and a notably higher amount of preserved tissue. Examination of supra- and propriospinal connections indicated an increased number of connections between the segments below the lesion and different cranial regions of the CNS following treatment. Astrocytes, microglial cells, and neurons were found to express each therapeutic protein for up to 5 days after mRNA-LNP injection in the injured spinal cord. These findings suggest that delayed treatment with mRNA-LNPs encoding a combination of therapeutic proteins can promote both structural and functional recovery after spinal cord contusion [158]. The intraspinal mRNA administration is a minimally invasive method of treatment in experimental animals with potential for further refinement [10]. Fig. 4B shows the effect of mRNA-based LNP treatment on spinal cord injury.

3.6. Ischemic stroke

Ischemic stroke (IS) is a prevalent neurological condition resulting from the constriction or blockage of blood vessels supplying the brain. The global occurrence of ischemic stroke is increasing due to the aging population, making its management a major public health concern [159,160].

3.6.1. Manifestation and pathophysiology of ischemic stroke

The sequence of molecular processes triggered by acute focal ischemia can be outlined as a time-sensitive series, marked by reduced energy generation (through the shift to anaerobic glycolysis during ischemia); hyperactivation of neuronal glutamate receptors (excitotoxicity); abnormal buildup of sodium, chloride, and calcium ions within neurons; damage to mitochondria; and eventual neuronal demise [161]. During the initial phase of a stroke, microglia play a crucial role in responding to the ischemic insult, aiding in recovery by clearing debris and producing anti-inflammatory mediators. As the condition progresses, the initially recruited protective microglia undergo a phenotypic shift from protective M2-dominant to pro-inflammatory M1-dominant, leading to an increase in inflammatory substances like cytokines, chemokines, and ROS. This shift contributes to the development of neuroinflammation, BBB breakdown, and neuronal death. Targeting molecules that regulate this shift presents an attractive therapeutic intervention to reduce neuroinflammation and maintain BBB integrity post- stroke [148,162].

IL-10 can drive microglia polarization towards the M2 phenotype through specific signaling pathways, thereby resolving neuroinflammation, protecting the BBB, and restoring neurological function post-stroke [163]. Developing strategies for local IL-10 expression in the ischemic brain is crucial to promote microglial polarization towards M2 phenotypes for mitigating neuroinflammation and preserving BBB post-stroke [164].

On the other hand, the ectopic overexpression of the transcription factor Olig2 in neural stem cells also triggers the initiation of developmental oligodendrogenesis is and promotes irreversible commitment to the oligodendrocyte lineage. In a rodent experiment, the activation of Olig2 stimulated the initial formation and differentiation of CNS progenitors into oligodendrocytes, resulting in increased and expedited

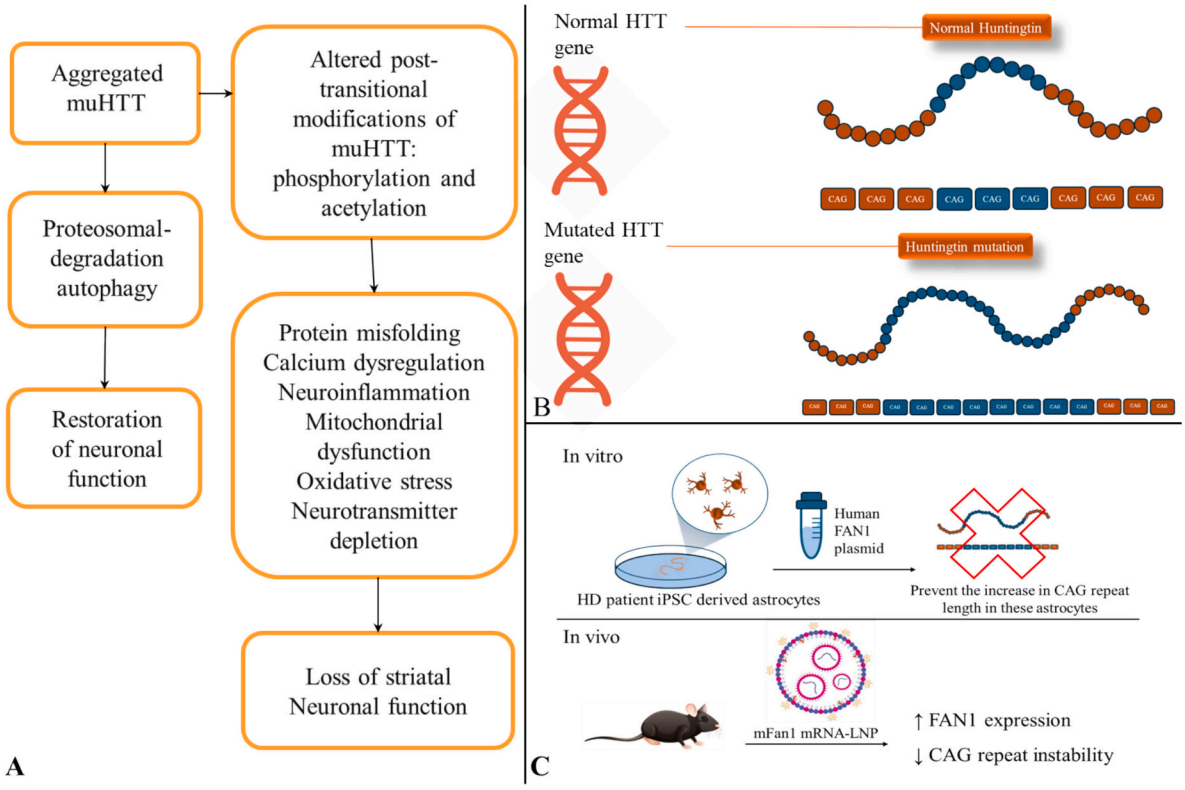


Fig. 6. Schematic representation of (A) Huntington's disease pathophysiology: it suggests that aggregation of muHTT can promote post-transitional modifications, which in turn cause loss of neuronal function and (B) The comparison of healthy and diseased HTT gene (C) The proposed treatment: In vitro experiments indicate that the overexpression of FAN1 protein can be prevented by multiple transfections with codon-optimized mFAN1 mRNA-LNP. In vivo study has shown that HTT CAG repeat instability is prevented post mFAN1 mRNA-LNP injection. Enhancing FAN1 protein expression could have a protective effect in mitigating expanded somatic HTT CAG repeats.

myelination in the corpus callosum [159,165]. The specific deletion of Olig2 during the multipotent progenitor stage led to the generation of oligodendrocyte precursor cells (OPCs), while impeding the expression of genes related to myelin. This suggests that while Olig2 functionality may not be essential for the differentiation of progenitor cells into OPCs, it is crucial for the maturation of OPCs into oligodendrocytes [165]. Fig. 5A shows the main pathophysiology suggested for ischemic stroke.

3.6.2. *Ischemic stroke therapy*

The primary objectives of IS treatment involve promptly reinstating regular cerebral blood flow and safeguarding neurons by halting or delaying the progression of ischemic events [161]. To date, the only approved medication for the treatment of acute ischemic stroke (AIS) is intravenous thrombolytic alteplase. Intravenous thrombolysis can be administered alone or in combination with endovascular treatment involving mechanical thrombectomy, particularly in cases of large vessel occlusion and within 6 h from the onset of symptoms. The potential risk of bleeding complications, particularly symptomatic intracerebral hemorrhage, has led to hesitancy in administering Intravenous thrombolysis. However, Tenecteplase, an alternative fibrinolytic agent, shows promise as it possesses a better safety profile compared to alteplase [166]. Nonetheless, despite numerous failed drug trials for AIS, one of the commonly cited reasons for these failures is the inability of drugs to reach sufficient concentrations in the at-risk penumbra. In order to address this challenge, the utilization of nanotechnology platforms to significantly enhance drug concentration in the BBB of the penumbra holds great potential [167].

The compromised BBB during IS remains a significant obstacle for drug penetration, leading to therapeutics failing to reach the desired concentrations and hindering therapeutic outcomes. Consequently, there has been a growing focus on the development of nanosized drug delivery systems, which aimed to enhance the therapeutic effects of IS-related therapeutics by facilitating increased drug accumulation in the ischemic brain [160,168].

A well-designed distribution mechanism should consider factors such as the specific cytokines needed for ischemic recovery, the optimal dosage and timing of cytokine delivery, and the most effective route of administration. To address this unmet need, Gao's research team utilized mRNA encoding phenotype-switching interleukin-10 (mIL-10) loaded in M2 microglia-targeting LNPs (mIL-10@MLNPs) to target ischemic regions in a mouse model of IS induced by transient middle cerebral artery occlusion. Their study demonstrated that intravenously administered mIL-10@MLNPs stimulated IL-10 production and promoted M2 (protective phenotype of microglia) polarization of microglia. This positive feedback loop contributed to resolving neuroinflammation, restoring the compromised BBB, and preventing neuronal apoptosis post-stroke as well as extending its effect at least 72 h post-stroke [162]. On the other hand, Nie's team focused on delivering oligodendrocyte-specific (Olig2) mRNA using LNPs and found that CD140a-targeted LNPs loaded with Olig2 mRNA (C-Olig2) facilitated the differentiation of PDGFRα^{+} OPCs derived from mouse neural stem cells into mature oligodendrocytes in vitro. Their results suggested that mRNA-mediated Olig2 overexpression could be a viable strategy to enhance oligodendrocyte differentiation and remyelination. Moreover, administration of C-Olig2 in a murine IS model resulted in improved BBB integrity, enhanced remyelination, and rescued cognitive deficits. Through a comprehensive analysis involving bulk RNA sequencing (RNA-seq) and single-nucleus RNA-seq (snRNA-seq), it was revealed that mice treated with C-Olig2 exhibited upregulated biological processes associated with learning and memory compared to those receiving empty LNPs [159].

A study done by another team demonstrated that conjugated LNPs that were bound with antibodies targeting vascular cellular adhesion

molecules (VCAM) expressed at the BBB were utilized in an animal model of large vessel AIS (considered as the gold standard AIS animal model) followed by mechanical thrombectomy (transient middle cerebral artery occlusion [tMCAO]) [167]. In both AIS in humans and mice tMCAO severe acute inflammation in the penumbra around the core infarct could be found, which all lead to the pregulation of VCAMs. In this study, these VCAM-targeting LNPs demonstrated significantly enhanced brain delivery, approximately two orders of magnitude higher compared to non-targeted conjugated lipid nanocarriers and ICAM (intercellular adhesion molecule-1) and the constitutively expressed PECAM (platelet endothelial CAM) as other targeting moieties. Furthermore, VCAM-targeted LNPs containing luciferase-encoding mRNA and Cre-recombinase exhibited selective and local expression in the ischemic brain region following intravenous administration. The administration of anti-inflammatory drugs post-IS resulted in a reduction of cerebral infarct volume by 62 % (IL-10 mRNA) or 35 % (dexamethasone) with 100 % survival rate only when these drugs were encapsulated within VCAM-targeted LNPs, showing that these LNP types presents a novel platform with therapeutic efficacy in the compromised BBB of the penumbra, and AIS [167]. The effect of VCAM and IL-10 mRNA-based LNP treatment on ischemic stroke is summarized in the Fig. 5B.

3.7. Huntington disease (HD)

Huntington's disease (HD) is inherited in an autosomal dominant manner and typically manifests in middle age, between 30 and 50 [169]. Repeat expansion disorders such as HD are caused by unstable repetitive sequences in the human genome (in the coding or non-coding regions), affecting the nervous system and leading to severe diseases. The clinical symptoms of repeat expansion disorders such as HD typically appear earlier in life and worsen over time during generations. Research has shown a strong correlation between the size of the repeat expansion and the age at which symptoms begin and/or the severity of the disorders [170].

3.7.1. Manifestation and pathophysiology of Huntington disease

HD is a dominant, monogenic, and inherited neurological disorder caused by the expansion of the genomic cytosine-adenine-guanine huntingtin (CAG) trinucleotide in the huntingtin gene, leading to an elongated polyglutamine segment in the Huntingtin (HTT) protein [171].

The Mutant HTT protein disrupts transcription processes, interferes with the normal functioning of mitochondria, and undergoes post-translational modifications (phosphorylation and acetylation) that contribute to complex pathogenic mechanisms such as proteasomal-degradation autophagy. Fig. 6B shows the mutation of HTT protein. The expansion of CAG repeats is evident in specific somatic and meiotic tissues, with the primary neurodegenerative consequences stemming from the extensive loss of neurons in the striatum and cortex tissues [172–174]. Despite the association between the length of HTT CAG repeats and the progression of HD, the disease onset can still vary by several decades among individuals with identical CAG repeat lengths [175].

One of the primary characteristics of HD is the presence of choreiform movements, which are involuntary, erratic, and writhing movements of the limbs, face, and trunk. Individuals with HD often struggle with coordination, balance, and speech [176], alongside cognitive issues like memory and attention loss, and impaired problem-solving and decision-making [177],depression, anxiety, irritability, mood swings, and personality changes, which can precede the onset of motor symptoms and greatly impact on the quality of life [178]. Fig. 6A shows the main pathophysiology suggested for Huntington's disease.

3.7.2. Huntington therapy

Therapeutic strategies primarily focus on managing the symptoms associated with the disease. The current treatment regimens aim to address motor, cognitive, and psychiatric symptoms with the goal of enhancing the quality of life for individuals living with HD [179].

One approach involves the use of dopamine modulators such as Tetrabenazine (TBZ) and Deutetrabenazine, both of which are approved by the US Food and Drug Administration (FDA) for treating chorea in HD [180]. These medications work by inhibiting the vesicular monoamine transporter (VMAT2), thereby reducing the levels of certain neurotransmitters [181]. Neuroleptics, conversely, function by blocking dopamine (DA) neurotransmission. They have the potential to alleviate symptoms of chorea, agitation, and psychosis [181]. Anti-glutamatergic drugs like amantadine, an antagonist to the *N*-methyl-D-aspartate (NMDA), may also be used for HD treatment Amantadine may counteract excessive glutamate neurotransmission in the basal ganglia, which helps reduce NMDA receptor-mediated excitotoxicity and the subsequent degeneration of striatal neurons in the HD brain. Furthermore, amantadine also affects dopaminergic synapses, while its exact mechanism is not well understood [182]. Recent advancements in gene therapy offer new hope for the Huntington's community. Retrospectively it is proposed that FAN1 gene (modifies HD progression by stabilizing HTT CAG repeat) and MMR (mismatch repair, a key gene in development of HD) components may influence homology-directed DNA damage repair through a shared mechanism [175].

To further investigate the potential therapeutic role of FAN1, Cheng and his team utilized the mRNA-LNP modality. By delivering codon-optimized FAN1 mRNA using LNPs, the researchers aimed to increase FAN1 protein levels and assess its impact on CAG repeat length in FAN1 knockout HD mouse model. They conducted their experiments using astrocytes derived from induced pluripotent stem cells (iPSCs) of HD patients, providing a valuable model for studying the effects of FAN1 on CAG repeat expansion. The findings revealed that the delivery of codon-optimized FAN1 mRNA-LNP effectively prevented the increase in CAG repeat length in these astrocytes. This suggests that enhancing FAN1 protein expression could have a protective effect in mitigating expanded somatic HTT CAG repeats. Overall, these findings highlight the potential therapeutic role of FAN1 in addressing repeat expansion disorders such as HD. Increasing FAN1 protein levels may reduce CAG repeat instability and slow down the progression of the disease. The use of mRNA-LNP delivery systems offers a promising approach for enhancing FAN1 expression (showing the role of FAN1 in maintaining genomic stability) and could pave the way for future therapeutic strategies targeting repeat expansion disorders [56]. The effect of mFan1 mRNA-LNP treatment is summarized in Fig. 6C.

3.8. Friedreich's ataxia

Friedreich ataxia (FRDA) is an uncommon hereditary movement disorder and a severe multisystemic disorder. Typically, FRDA manifests during adolescence with gradual and progressive instability, dysmetria, and dysarthria, ultimately resulting in the loss of independent walking ability and severe disabilities [183,184].

3.8.1. Manifestation and pathophysiology of Friedreich's ataxia

FRDA primarily affects the sensory pathways and is accompanied by sensory loss and abnormalities in deep tendon reflexes, along with non-neurological symptoms like cardiomyopathy, diabetes, and skeletal abnormalities like scoliosis and pes cavus [183].

Approximately 96 % of patients with FRDA experience its onset due to the presence of biallelic GAA-trinucleotide expansions in the first intron of the frataxin gene. The intronic GAA expansions have the effect of suppressing the transcription of the frataxin gene, resulting in decreased levels of a protein that is otherwise normal [184]. In individuals with FRDA, reduced levels of FXN protein have been observed across various cell types. Interestingly, while complete elimination of FXN is fatal, the pathology associated with diminished levels of FXN is specific to certain cell types such as neurons, cardiomyocytes, and

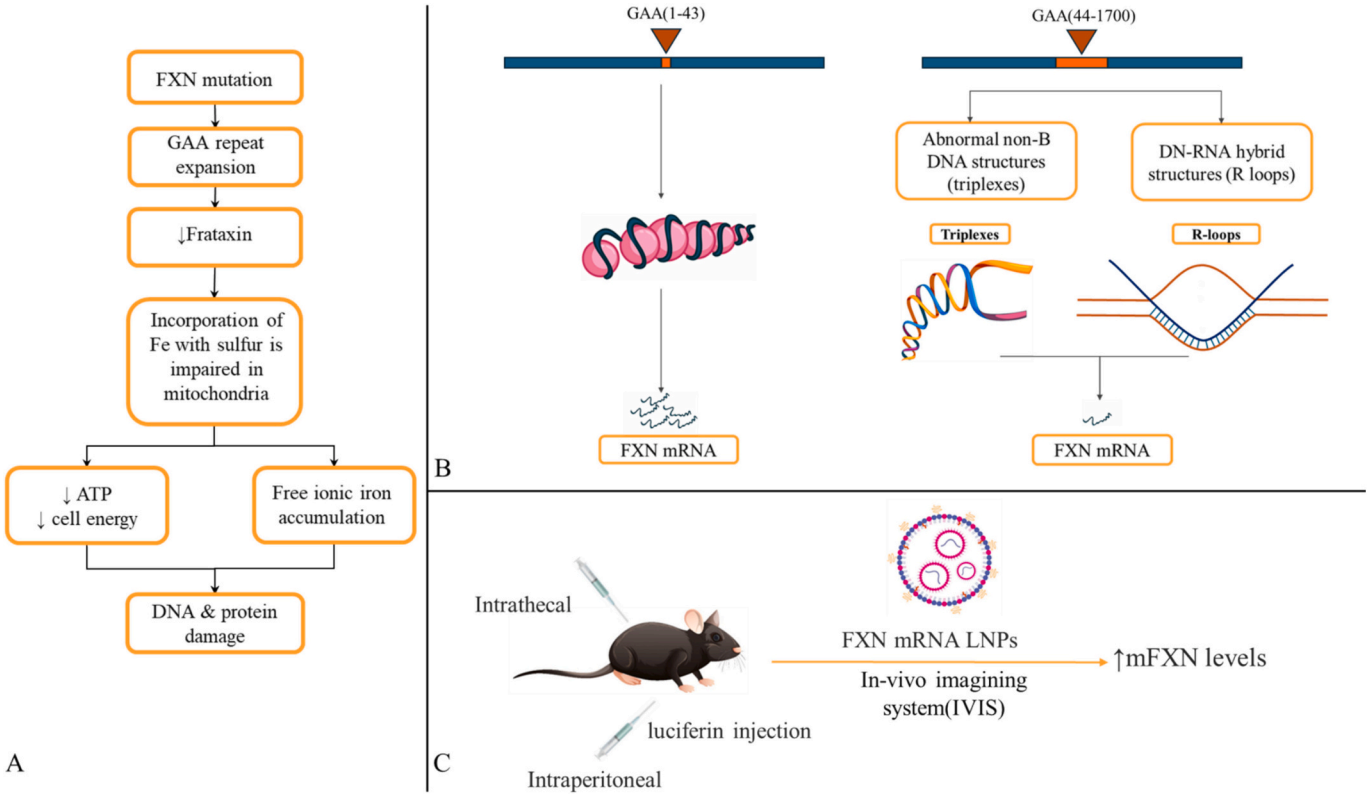


Fig. 7. Schematic representation of (A) ischemic Fredrich's ataxia pathophysiology. FXN mutation can cause DNA damage and trigger FRDA (B) The difference between unaffected and FRDA patients in GAA-TTC repeats: unaffected individuals who carry up to 43 GAA-TTC repeats, contain active histone marks of gene transcription initiation and elongation at the FXN promoter and intron 1 regions. In FRDA patients, the presence of large GAA-TTC repeat expansion leads to FXN gene silencing by two potential mechanisms: the GAA-TTC repeat may adopt abnormal non-B DNA structures (triplexes) or DNA-RNA hybrid structures (R loops), which impede the process of RNA polymerase and thus reduce FXN gene transcription (C) FRDA proposed treatment: effect of FXN mRNA LNP on frataxin in mice.

pancreatic islets. In the context of the CNS, the progressive degeneration caused by FXN deficiency leads to disease. Therefore, therapies that can effectively target the CNS are highly sought after [184,185]. Fig. 7A summarizes the main pathophysiology suggested for FRDA while Fig. 7B suggests the effect of GAA repetition of FRDA pathophysiology.

3.8.2. *Friedreich ataxia therapy*

Current treatment strategies being considered involve reducing oxidative stress, enhancing cellular frataxin levels, optimizing mitochondrial function, and regulating metabolic pathways controlled by frataxin. Additionally, physical therapies are being investigated to improve functionality in individuals with FRDA. Antioxidants like Idebenone [186], Coenzyme Q10 [186,187], and Thiamine [188], as well as anti-inflammatory therapies such as methylprednisolone and Deferiprone [189], may be beneficial in alleviating symptoms of FRDA. It is important to note that there is currently no approved pharmacological treatment for this condition. Despite the lack of approved pharmacological treatments, research has made significant progress since the genetic basis of FRDA was identified, indicating promising developments [190]. Various molecules, including small molecules like histone deacetylase inhibitors and nicotinamide, and large molecules such as engineered transcription activator-like effectors, are being investigated to upregulate FXN in FRDA therapy. Additionally, strategies such as FXN protein supplementation and viral gene transfer are being explored as potential therapeutic options [191]. One promising approach currently being pursued is RNA transcript therapy (RTT), which involves encapsulating exogenous mRNA molecules in LNPs for replacement or supplementation purposes. RTT has shown promise in multiple therapeutic applications and has attracted significant investments in biopharmaceutical companies [185]. Proof-of-principle studies have demonstrated the successful rescue of a lethal genetic knockout mouse model and the delivery of therapeutic proteins in disease models using RTT. These findings suggest that RTT could be a viable replacement approach for inherited recessive diseases like FRDA. Researchers have investigated the use of intravenous and intrathecal LNP based delivery of recombinant human FXN (hFXN) mRNA to explore the concept of RTT. The study provided evidence that hFXN mRNA can be effectively translated in vitro through cellular transfection and in vivo when administered systemically. Furthermore, it was observed that the resulting protein, known as mature functional form (mFXN), undergoes proper processing. Notably, the study revealed that the maturation machinery is not a limiting factor, and the in vivo half-life of mFXN protein exceeds one week, indicating its stability. Additionally, the researchers tested the intrathecal delivery method and observed the uptake of hFXN mRNA in the dorsal root ganglia, which affects primary site of pathology in FRDA. These findings highlight the potential of LNP-based delivery of hFXN mRNA as a supplementary therapy for treating FRDA and other CNS disorders where dorsal root ganglia are involved [185]. The effect of FXN mRNA LNP in vitro and in vivo treatment is summarized in Fig. 7C.

3.9. *Alzheimer's disease*

Alzheimer's Disease (AD) is a major cause of dementia worldwide, impacting not only patients but also families, society and the economy [192], with genetic factors playing a significant role in its heritability and over 40 genetic risk loci identified, most notably the APOE alleles [193,194].

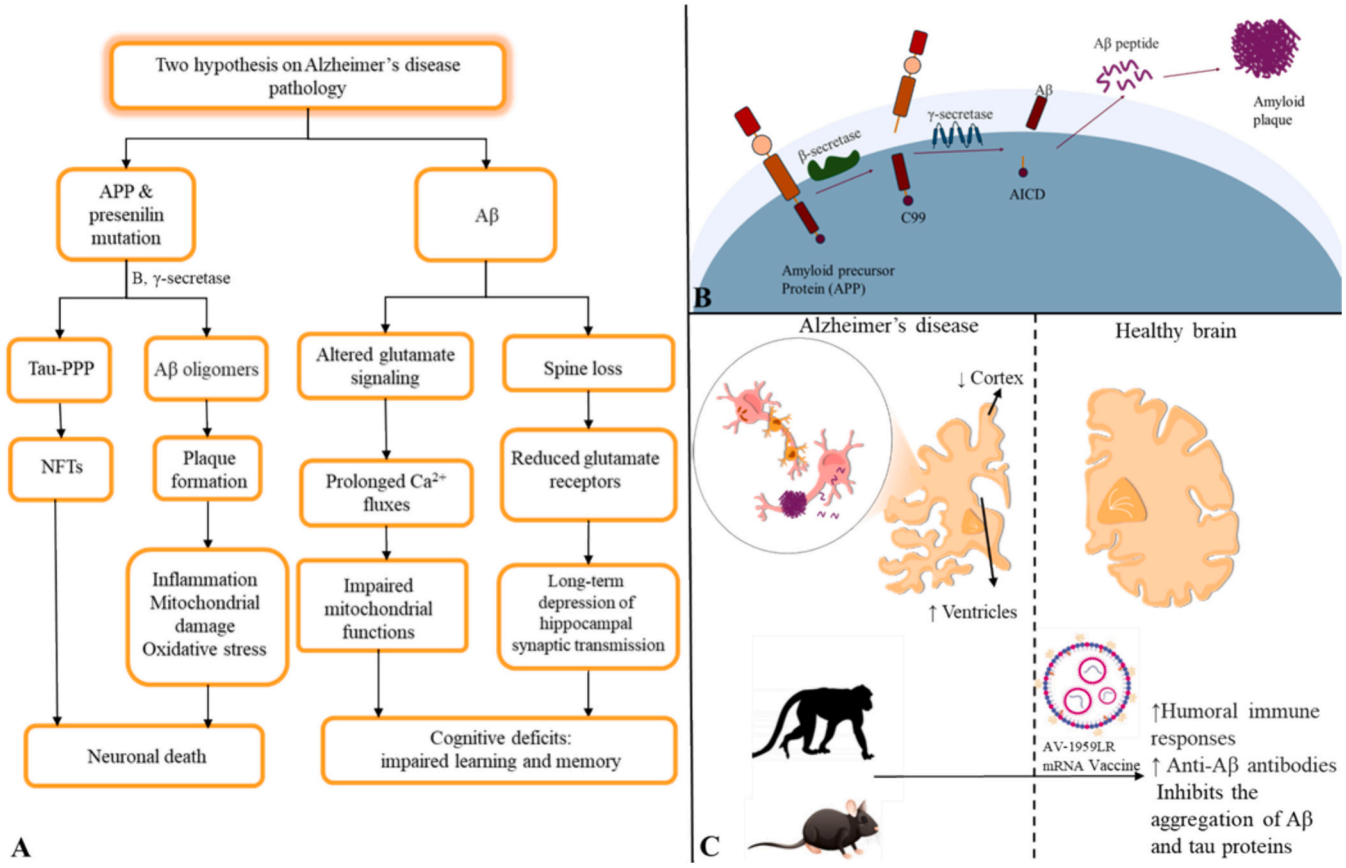


Fig. 8. Schematic representation of (A) Alzheimer's disease pathophysiology, (B) Amyloid plaque formation: the formation of the plaque results from the dysregulation of amyloid precursor protein (APP) processing. Amyloid-beta (Aβ) proteins are synthesized within neuronal membranes and subsequently aggregate in the extracellular environment, leading to the formation of Aβ plaques. These plaques disrupt neuronal function and initiate inflammatory responses. Concurrently, neurofibrillary tangles arise from the accumulation of hyperphosphorylated tau proteins within neurons, which displace organelles and hinder the vesicular transport essential for synaptic transmission. The interplay of these two pathological processes is associated with neuronal cell death and brain atrophy, contributing to the gradual decline in cognitive abilities. (C) Proposed treatment for AD: LNP-based platform of mRNA containing AV-1959 antigen (called AV-1959LR) to target the N-terminus of pathological Aβ in mice and non-human primates to induce humoral responses, stimulate antibodies, and inhibit the aggregation of Aβ and tau proteins.

3.9.1. Manifestation and pathophysiology of Alzheimer's disease

APOE polymorphic alleles serve as the primary genetic factors influencing the risk AD. Individuals who possess the ε4 allele exhibit a heightened susceptibility to AD in contrast to those with the more prevalent ε3 allele, while the ε2 allele is linked to a reduced risk. Furthermore, the presence of the APOE ε4 allele correlates with an elevated risk of cerebral amyloid angiopathy and cognitive decline associated with aging in the absence of pathological conditions [194]. The Amyloid-β (Aβ) peptide (molecular weight = 4 kDa), is a fragment derived from the amyloid precursor protein (APP). APP is a larger precursor molecule that is widely produced by various cells in the brain, including neurons, vascular and blood cells (including platelets), and to a lesser extent, astrocytes. The generation of Aβ involves two consecutive proteolytic cleavages of APP [195]. The conformation of soluble Aβ transforms to a high β-sheet composition, making it more likely to aggregate into polymeric structures like soluble oligomers and larger insoluble fibrils. Over time, these fibrils deposit as extracellular amyloid plaques in the brains of individuals with AD [196]. The degradation of Aβ is a physiological process facilitated by peptidases such as insulin-degrading enzyme, neprilysin, and endothelin-converting enzyme. Aβ can be eliminated from the brain through multiple pathways, including transportation across the BBB, interstitial fluid bulk flow, arachnoid villi, and glymphatic-lymphatic pathways. Furthermore, Aβ aggregates can be engulfed and degraded by microglia, perivascular macrophages, and astrocytes. Any disruption in these clearance mechanisms can lead to an imbalance between Aβ production and clearance in the brain, ultimately resulting in neuronal dysfunction and neurodegeneration [197,198].

It has been found that an imbalance in Aβ neuronal production and extracellular Aβ clearance may serve as the initiating factor for Aβ dyshomeostasis leading to protein misfolding, aggregation, and the early accumulation of Aβ in plaques followed by γ-secretase cleavage at intramembranous sites, resulting in the production of Aβ [195,199]. Tau proteins can also be aggregated in separate neuronal networks in AD patients. The abnormal accumulation of Aβ and hyperphosphorylated tau proteins are the two main neuropathological characteristics in AD. The mechanism of plaque formation is summarized in Fig. 8B. These pathologies affect distinct neuronal and glial systems, ultimately leading to neurodegeneration and cognitive decline in individuals with AD [200]. Fig. 8A shows the main pathophysiology suggested for Alzheimer's disease.

3.9.2. Alzheimer's Disease therapy

The combination therapy of memantine and Cholinesterase inhibitors (ChEIs) has emerged as the most extensively studied pharmacological approach for treating AD [201], showing clinical efficacy in managing the AD-associated symptoms. Initially, tacrine, a first-generation ChEI, was employed; however, its use was limited due to the occurrence of hepatotoxic side effects. Subsequently, donepezil, rivastigmine, and galantamine were introduced, with donepezil being the most widely used agent. The effectiveness of these different agents appears to be similar, thus the choice of treatment should be based on

Table 1
The LNP-mediated mRNA delivery for treatment of different neurodegenerative diseases.

Disease	LNP component	mRNA type	Rout of administration	Model	Outcome	Ref
Parkinson's disease	DLin-MC3-DMA, DSPC, Cholesterol, PEG-lipid (50:10:38.5:1.5)	GM-CSF mRNA	_	In vivo (C57BL/6 mice & male Sprague-Dawley (SASCO) rats)	↑ Plasma GM-CSF levels nigrostriatal neuroprotection ↓ Reactive microgliosis ↑ Treg and myeloid populations	[122]
	DLin-MC3-DMA, DSPC, Cholesterol, PEG-lipid (50:10:38.5:1.5)	GM-CSF mRNA	Stereotactic injection	In vivo (C57BL/6 mice & male Sprague-Dawley (SASCO) rats)	↑ Plasma GM-CSF levels (dose-dependent) ↑ Treg ↑GM-CSF bioavailability Extension of GM-CSF dosing and duration of action	[125]
Multiple sclerosis	COATSOME® SS-OP18:2 PS, Cholesterol, DMG-PEG2000 (52.5/7.5/40/1)	$MOG_{27–63}$-expressing mRNA	Intravenous	In vivo- (Female C57BL6/J mice)	Tolerance induction ↓EAE symptoms ↑ Treg ↓$CD4^+/CD8^+$ ratio	[137]
	Passive immune system targeting by PS Ionizable amino lipid (heptadecan-9-yl 8-((2-hydroxyethyl) (8- (nonyloxy)-8-oxooctyl)amino)octanoate / 2-(dinonylamino)-1- (4-(N-(2-(dinonylamino) ethyl)-N-nonylglycyl)piperazin-1-yl) ethan-1-one), DSPC, Cholesterol, PEG-DMG (50:10:38.5:1.5)	ARG1 mRNA	Intraperitoneal	In vivo (male / female C57BL/6 arginase-deficient mice)	↑ Plasma ammonia ↑ Plasma arginine ↑ Plasma guanidino ↑Density of myelinated axons ↑ Thickness of the myelin sheath ↓ Associated leukodystrophy Restoring normal oligodendrocyte function	[135]
Spinal cord injury	L-319, DSPC, Cholesterol,PEG-DMG (55:10:32.5:2.5)	Human IL-10 (hIL-10) -encoding mRNA	Intraspinal	In vivo (Sprague-Dawley female rats)	Spinal cord injury repair Restoration of Motor Function ↓$CD6_8^+$ macrophages and Iba-1^+ microglia, IL-1b, ↓TNF- α, and CCL3	[10]
Ischemic stroke	LNP	Olig2 mRNA	Intravenous	In vitro & in vivo	↑ Remyelination in IS ↑ BBB integrity improves cognitive deficits	[159]
	PDGFRα/CD140 as targeting ligand AA3-ionizable lipid, Cholesterol, PEG-lipid DSPC, DMG-PEG, DOPE as helper lipids	mIL-10 mRNA	Intravenous	In vivo (transient middle cerebral artery occlusion (MCAO) mouse model)	↑ M_2 polarization of microglia ↓ BBB disruption	[162]
	M2 microglia as targeting DPPC, Cholesterol, DSPE-PEG2000-azide) (54:40:6) VCAM as targeting agent	luciferase-encoding mRNA and Cre-recombinase	Intravenous	In vivo (male C57BL/6 mice)	Higher delivery rate of VCAM targeted LNPs to brain 73 % ↓ in infarct volume 100 % survival rate with no bleeding or lung damage	[167]
Huntington's disease	ALC0315, Cholesterol, DSPC, ALC-0159 (46.3:42.7:9.4:1.6)	FAN1 mRNA	In vivo transfection	In vitro (iPSC-derived astrocytes) In vivo (male HD mouse model (HdhQ111))	Preventing CAG repeat expansion ↑ FAN1 expression ↓ CAG repeat instability	[56]
Friedreich's ataxia	DLin-MC3-DMA, DSPC, Chol, DMG-PEG2k (55:10:32.5:2.5)	Frataxin (FXN) mRNA	Intrathecal	In vitro and in vivo (female BALB/c mice)	First RNA transcript therapy (RTT) for delivery of therapeutic mRNA to dorsal root ganglia	[185]
Alzheimer's Disease	ALC-0315, Cholesterol, phospholipid, PEG-Lipid (50:38.5: 10:1.5)	AV-1959LR mRNA	Intramuscular	In vivo (female C57BL/6 mice (H-2b haplotype) / female cynomolgus monkeys (*Macaca fascicularis*)	Induction of robust humoral immune responses after a minimal dose of mRNA–LNP ↑ Anti-Aβ antibodies Inhibits the aggregation of Aβ and tau proteins Inhibits the aggregation of Aβ and tau proteins and delays the onset of dementia	[192]

considerations such as cost, individual patient tolerance, and the experience of the physician [202].

Passive immunotherapies with monoclonal antibodies (mAbs) and vaccines have emerged as the most reliable and promising strategies for AD treatment. Recent data from clinical trials involving FDA-approved mAbs, such as aducanumab and lecanemab, have provided solid evidence that targeting the N-terminal region of Aβ and clearing Aβ aggregates from the brains of individuals with early AD can slow down the

progression of the disease [203]. However, while both mAbs have received FDA approval, only lecanemab (Leqembi®) is currently being used for biweekly intravenous (IV) treatment at a dosage of 10 mg/kg/injection for individuals with early AD. Despite modest clinical benefits, Leqembi® highlights the importance of early intervention in AD's pathogenic cascade. Nevertheless, the practicality of using mAbs as a preventive measure is hindered by their complex nature, high cost, and the need for frequent high-dose intravenous administrations. Furthermore, there is a notable risk of amyloid-related imaging abnormalities, including cerebral microhemorrhages and edema [192].

Data from immunotherapy studies have also revealed that adequate concentrations of antibodies in the periphery of vaccinated individuals are crucial for effectively targeting Aβ in the CNS [192,204,205]. To address this, a vaccine platform called MultiTEP has been developed as the foundation for a DNA vaccine named AV-1959D. This vaccine specifically targets pathological Aβ and has successfully completed IND (Investigational New Drug) enabling studies. Subsequently, a Phase I clinical trial was initiated, enrolling early AD volunteers [206]. This group further developed a LNP-based platform of mRNA containing AV-1959 antigen (called AV-1959LR) composed of MultiTEP and three copies of a Aβ42 (42 amino acid form of Aβ), as an alternative to the DNA vaccine AV-1959D. In this current pilot study, the advanced MultiTEP technology was combined with a novel mRNA approach to develop AV-1959LR, an mRNA vaccine that is encapsulated in LNPs. This study showed the feasibility of mRNA –LNP based vaccine for AD in both mice and non-human primates that presented more robust immunogenicity, and lower manufacturing costs than DNA vaccine AV-1959D [192]. Therefore, vaccination strategies targeting pathological proteins could be developed using LNP-encapsulated mRNA to induce an immune response against these proteins. The effect of AV-1959LR vaccine in mouse and monkey is depicted in Fig. 8C.

LNPs offer an effective method for delivering mRNA therapeutics to the brain, where they can target specific proteins involved in NDD. Table 1 shows the LNP-mediated mRNA delivery for treatment of different neurodegenerative diseases including Parkinson's disease, Alzheimer's disease (AD), multiple sclerosis (MS), Huntington's disease (HD), ischemic stroke, spinal cord injury, and Friedreich's ataxia.

4. Strategies to improve the delivery and effectiveness of mRNA therapeutics

In order to improve the effectiveness of RNA therapeutics, it is crucial to devise innovative delivery methods to overcome the barriers associated with mRNA delivery, particularly the BBB, and improve stability, transfection efficiency, cellular uptake, and therapeutic efficacy. Here, we discuss different strategies to enhance the delivery and effectiveness of mRNA therapeutics.

4.1. Modifications of RNA

4.1.1. Chemical modifications of mRNA

Chemical and physical modifications of mRNA could be beneficial for enhancement of their efficacy [207,208]. Modifying the cap structure, 5′ and 3′UTRs, coding sequence, and poly(A) tail length can have a substantial impact on the efficacy of LNP-mRNA therapeutics [209]. Post-translational modifications of mRNA, in addition to the 5′ cap, 5' UTR, and 3' UTR and ORF modification, can hinder the immune recognition of mRNA. Minimizing the reorganization of mRNA by the innate immune system can prevent the detection and destruction of mRNA translation, leading to increased expression. One effective method to avoid immunogenicity is the removal of the uncapped phosphate of mRNA through the action of phosphatases. Furthermore, the integration of chemically altered nucleosides is a key approach for diminishing the immunogenicity of in vitro transcription (IVT) mRNA [210]. Typical chemical modification techniques consist of pseudouridine (Ψ), N6-methyladenosine (m6A), 5-methylcytidine (m5C), and N1-methylguanosine (m1A) [14]. Pseudouridine is a modified isomer of uridine that replaces the nitrogen-carbon glycosidic bond with a carbon-carbon bond. It has been shown to reduce the recognition of RNA by pattern recognition receptors (PRRs) such as Toll-like receptors (TLRs) and retinoic acid-inducible gene I (RIG-I). This modification prevents the activation of innate immune responses, including the production of pro-inflammatory cytokines and type I interferons. Incorporating pseudouridine into the RNA sequence enables LNP formulation to avoid triggering an immune response, allowing for efficient delivery of the therapeutic payload. Similarly, 2-thiouridine, a sulfur-modified nucleotide, has been found to inhibit immune activation by reducing RNA recognition by TLRs, enhancing RNA stability and reducing its degradation by nucleases, thereby improving the efficiency of delivery and minimizing immune activation. Additionally, the incorporation of 5-methyl cytidine, a modified form of cytidine, into the RNA sequence can further enhance the stability and reduce immunogenicity of the NA payload, reduce the activation of immune sensors, and increase RNA half-life in cells, leading to improved delivery and reduced immune activation [211,212].

By combining these chemically modified nucleotides within the LNP formulation, the immune activation associated with NA delivery can be effectively circumvented. This allows for the safe and efficient delivery of mRNA vaccines or gene therapies, without triggering unwanted immune responses [211]. Additionally, enhancing RNA purity and the eliminating impurities are crucial in preventing mRNA degradation and immunogenicity in mRNA cancer vaccines [213,214]. Therefore, incorporating modified nucleotides into the mRNA to improve stability, translation efficiency, and resistance to degradation can enhance the overall effectiveness of the mRNA-LNP formulation.

4.1.2. Advanced RNA modalities

4.1.2.1. Self-amplifying RNAs (saRNAs). A major limitation of traditional mRNA in gene therapy is its transient expression, which reduces therapeutic effectiveness. To overcome this, incorporating multimeric mRNAs can enhance target gene expression [215]. Several approaches have been proposed for generating multimeric RNAs, including chemical conjugation (cross-linked RNA), self-assembled RNA, enzymatically elongated RNA, and self-amplifying RNA (saRNA) [216]. Notably, innovative strategies like saRNA and circular RNA offer promising ways to extend the effects of mRNA-LNPs [35].

saRNA is a cutting-edge RNA technology that combines the benefits of mRNA with self-replication, enabling prolonged and more efficient protein expression at lower doses. It contains an RNA sequence derived from alphaviruses (or other RNA viruses) that encodes an RNA-dependent RNA polymerase. Once inside the cell, this enzyme is produced, allowing continuous replication of the saRNA and amplifying protein production. This self-replication mechanism sustains protein expression for weeks or even months, significantly enhancing therapeutic effects while reducing required doses and potential side effects [217–219].

Due to its large size (~9500 nucleotides) and negative charge, self-amplifying RNA (saRNA) requires a delivery system for efficient cellular uptake and protection from degradation. LNPs containing cationic lipids shield saRNA from RNase degradation, even when attached to the particle's surface [220].

Notably, saRNA encapsulated in LNPs has demonstrated greater effectiveness compared to other nanoparticle formulations. For instance, an saRNA-LNP formulation designed to stimulate IL-6 production significantly enhanced immunogenicity, triggering stronger antibody and cellular immune responses against influenza and SARS-CoV-2 antigens compared to polymer-based nanoparticles [221]. In vaccine development, saRNAs also offer significant advantages as their prolonged antigen expression can strengthen immune responses. Preclinical studies have shown that saRNA-based COVID-19 vaccines induce long-

lasting antibody and T-cell responses, highlighting their potential for future vaccine strategies [219]. In addition to cancer and infectious diseases [217], saRNA vaccine also offer promising potential for preventing and treating neurodegenerative conditions. The use of saRNA-LNP platforms holds potential for treating various NDD, by enabling the possibility of reduced dosing and sustained expression of therapeutic proteins and which is crucial for addressing chronic conditions like neurodegenerative diseases

4.1.2.2. Circular RNAs. Innovative approaches like circular RNAs offer promising strategies to prolong the effect of RNA therapeutics [35]. Circular RNAs are covalently closed RNA molecules that lack free 5′ and 3′ ends, making them highly resistant to exonuclease-mediated degradation. This structural stability allows them to exhibit prolonged expression compared to linear mRNAs. Engineered with internal ribosome entry sites, circular RNAs enable cap-independent translation, leading to sustained protein production [222].

It is worth mentioning that recent studies suggest circular RNAs play a critical role in NDD onset and progression, making them potential diagnostic, prognostic, and therapeutic targets [223]. Circular RNAs are highly expressed in neuronal tissues and play a crucial role in regulating gene expression related to NDD. They act as microRNA sponges, influence transcription, and interact with RNA-binding proteins, thereby affecting neuronal function. Certain circular RNAs have been associated with the development of Alzheimer's and Parkinson's diseases, highlighting their potential as therapeutic targets [224].

Integrating circular RNAs into LNPs could enhance the efficacy of RNA therapeutics for NDD through sustained protein production. Preclinical studies have shown that circular RNA-LNPs can achieve protein expression lasting several weeks, whereas linear mRNA typically lasts only a few days [225]. Additionally, circular RNAs are compatible with LNP-mediated delivery, ensuring efficient translation without triggering RNA-mediated innate immune responses [226]. Moreover, circular RNA-LNP formulations have demonstrated higher thermal stability than mRNA-LNP vaccines, maintaining integrity at 4 °C for extended periods [227].

LNPs incorporating circular RNA have demonstrated significant therapeutic potential, including enhanced axon regeneration and neuroprotection. Notably, they have successfully delivered nerve growth factors to retinal ganglion cells, promoting robust neuroprotective effects [34].

4.2. Optimization of LNP components

The ability to traverse the BBB using various nanoparticles has demonstrated potential for the delivery of diverse therapeutic agents into the CNS [228]. However, achieving BBB permeability often necessitates complex modifications to the nanoparticles. A straightforward and efficient method for transporting therapeutic agents into the brain involves using synthetic lipids derived from neurotransmitters (NTs), which can transport small molecules, NAs, and proteins through a unified nanoparticle design. NTs, which are naturally occurring substances that facilitate neurotransmission, have been shown to possess the capability to penetrate the BBB [228]. For instance, dimethyltryptamine and certain derivatives of tryptamine have been identified as capable of crossing the BBB via active transport mechanisms across the endothelial plasma membrane [229]. Furthermore, tryptamine-derived lipidoids (referred to as NT1-lipidoids) have been effectively utilized to transport fluorescent dyes across the BBB without the need for additional targeting ligands. When incorporated into lipid formulations that are otherwise impermeable to the BBB, NT1-lipidoids enable the resultant LNPs (NT1-LNPs) to acquire the necessary BBB-penetrating properties [228].

Tanakas team developed a novel material for non-viral gene therapy known as SS-cleavable proton-activated lipid-like material. This material consists of three key structural components: hydrophobic scaffolds for bilayer formation, tertiary amines for proton absorption, and disulfide bonding for cytoplasmic cleavage. The gene-encapsulating LNP made from ssPalm (LNPssPalm) is neutrally charged under physiological conditions. This neutral surface property is advantageous for regulating pharmacokinetics and intracellular trafficking compared to commonly used cationic materials in gene delivery. The neutral carrier circulates systemically in a highly dispersed state, minimizing nonspecific interactions with endogenous macromolecules. Additionally, neutral gene carriers are generally less toxic. Upon cellular uptake, LNPssPalm acquires a positive charge due to the protonation of tertiary amines in the acidic endosomal environment. This positive charge facilitates interaction between the carriers and the endosomal membrane, promoting endosomal escape [230].

Tailoring the lipid composition (varying the type and ratios of lipids) can enhance stability and optimize the release profile of mRNA. It has been shown that MC3-based LNPs exhibit significant promise for the delivery of mRNA within the CNS. The administration of Cas9 mRNA/Ai9 sgRNA via MC3 LNPs facilitates effective gene editing within the striatum and hippocampus of Ai9 mice. The efficiency of local transfection achieved with MC3 LNP Cas9 mRNA/Ai9 sgRNA is comparable to or exceeds that of previously reported outcomes utilizing Cas9 ribonucleoproteins (RNPs). This finding suggests that MC3 LNPs are capable of transporting mRNAs encoding larger proteins, such as Cas9, in to the multiple brain cell types (neurons, astrocytes and microglia) by its direct intracerebral injections [231]. The respected Cre mRNA transfection was successful in the striatum and hippocampus with ~52 and ~ 49 % efficiency, respectively. MC3-based LNP Cas9 mRNA/Ai9 sgRNA edition rate in the striatum and hippocampus were ~ 7 % and ~ 3 %, respectively [231].

Based on their characteristics other ILs such as L-319 [232], ALC-0315 [192], Coatsome® SS-OP [137], and DLin-MC3-DMA (MC3) [55] can be used in LNPs to mitigate CNS diseases [159,162,233]. Interestingly, Mitchell's group utilized combinatorial chemistry to enhance the biodegradability and efficacy of ionizable lipids through structural optimization (in tail, head group, and linker) for mRNA delivery applications [234]. Collectively, it has been shown that ILs with improved endosomal escape properties can enhance the delivery of larger saRNAs or circular RNAs [49].

In addition, LNP formulations for mRNA transfection engineered with various ratios of PEG-lipid anchors and different anchor lengths (DSPE (C18)-PEG and DMG (C14)-PEG). The study showed that increasing long anchor PEG-lipid content resulted in a longer circulation time after intravenous administration in a mouse model of traumatic brain injury. However, excessive PEGylation can reduce cellular uptake, so the degree of PEGylation must be optimize. This PEG decoration can cause enhanced mRNA expression in injured brain tissue, which is indicative of LNPs gene expression with greater duration in the brain than off-target organs [235].

4.3. Functionalization and conjugation with targeting ligand

Functionalization of LNPs with targeting ligand can enhance site-specific delivery, uptake, and reduce off-target effects. LNPs have been modified with targeting ligand including antibodies to deliver drugs to non-CNS sites such as tumors, myeloid cells, and hepatocytes [236]. It is offered that receptor-specific monoclonal antibodies could be coupled to the surface of LNPs to enhance delivery to the brain. The monoclonal antibodies could facilitate receptor-mediated transcytosis of the LNP across the BBB via specific brain endothelial receptors, such as the insulin receptor (INSR), transferrin receptor (TfR) [237], and glutathione [27].

Besides antibodies, functionalization of nanoparticles with Apolipoprotein E (ApoE) has been reported to be involved in the brain uptake mechanism [238]. There have been accounts of effective transportation of mRNA to neuronal cells through the functionalization of LNPs with ApoE by targeting neuronal cells and facilitating delivery to the CNS

[230]. ApoE adsorption onto LNP surfaces is crucial for their uptake into liver cells/hepatocytes, leading to the formation of ApoE-LNP complexes that bind to lipoprotein receptors on hepatocytes. This enables endocytosis and accumulation of the LNPs in the liver, making them naturally directed to this organ [236]. Additionally, advancements in nanoparticle engineering have made it possible to create targeted and controlled-release delivery systems that can optimize the distribution and stability of RNA therapeutics.

To achieve the brain specific delivery, Mitchell's group designed a platform of peptide functionalized LNP (pLNP) for delivery of mRNA to the brain. They decorated the LNP with different peptides including RVG29 (binds to nicotinic acetylcholine receptors), AP2 (binds to LRP-1 receptors), T7 (bids to transferrin receptors), and mApoE (binds to LDLR receptors). Therefore, pLNPs are able to target neurons and brain endothelial cells with the overexpressed receptors. This platform showed enhanced mRNA transfection efficiency and improved BBB endocytosis and could be considered as a potential for treatment of neurological diseases [26]. These systems can be designed to protect RNA from degradation, facilitate cellular uptake, and release the therapeutic payload at specific sites or in response to particular stimuli.

Nanobodies, as single-domain antibodies, provide significant benefits for brain delivery due to their small size, high affinity, and low immunogenicity, enabling them to cross the BBB effectively through receptor-mediated transcytosis. Their ability to be engineered for improved pharmacokinetics and targeted delivery makes them promising candidates for both diagnosing and treating brain diseases such as Alzheimer's and Parkinson's diseases [24]. Single-domain nanobodies, provide innovative solutions for delivering and targeting immunotherapeutics in the brain and nervous system offering advantages over nucleic acid drugs by targeting modified proteins with high specificity and controllable binding affinities. Their specificity and engineering potential are enhanced when their binding sites are expressed intracellularly [239].

In this context, one study presented two novel nanobodies that specifically target amyloid deposits and tau neurofibrillary tangles, the hallmarks of Alzheimer's disease (AD). In transgenic AD mouse models, intravenous administration followed by real-time two-photon microscopy revealed gradual BBB extravasation, parenchymal diffusion, and effective labeling of pathological lesions. These findings highlight nanobodies as BBB-permeable probes for both extracellular and intracellular targets, offering new diagnostic and therapeutic possibilities in neurology [240]. In another study, an engineered transferrin receptor-binding nanobody (M1R56H, P96H, Y102H) with pH sensitivity has been developed to facilitate the transcytosis of biological macromolecules into the brain [25].

Targeting α-synuclein aggregation with nanobodies offers a promising strategy for slowing Parkinson's disease progression. The nanobodies VH14PEST and NbSyn87PEST significantly reduced phosphorylated α-synuclein levels, with VH14PEST demonstrating superior dopaminergic tone maintenance and reduced microglial inflammation. These findings highlight nanobodies as potential therapeutic agents for synucleinopathies like Parkinson's disease [241].

Another study introduced a novel anti- soluble amyloid-β oligomers (SAβOs) nanobody. SAβOs accumulate early (before plaque formation) and have a direct role in in Alzheimer's disease neurodegeneration. Nanobody crossed the BBB and effectively targeted amyloid species in 5XFAD mice. Furthermore, its ability to target pathological amyloid species suggested possible theragnostic applications and could serve both diagnostic and therapeutic roles [242].

A novel approach combining anti-GFAP (glial fibrillary acidic protein) nanobodies with gold nanoparticles and X-ray phase contrast tomography (XPCT) enabled 3D visualization of specific cells in NDD. By utilizing metal nanoparticles for X-ray attenuation and nanobody-based targeting, this technique enhances XPCT specificity, offering a promising molecular imaging method, particularly for studying astrocytes in mouse brains [243].

Very recently, BBB-crossing LNPs (BLNPs) have been designed by using small-molecule as ligand to deliver mRNA to the CNS. Through innovative design, the research team successfully synthesized 72 distinct types of BBB-crossing lipids by conjugating small-molecule ligand, such as L-DOPA, D-serine, and MK-0752, with amino lipids. This process produced BLNPs that demonstrate efficient mRNA encapsulation, achieving efficiencies ranging from 64.9 % to 88.1 %. In vivo screening has identified MK16 BLNPs as the leading formulation, achieving an 8.3-fold enhancement in mRNA delivery to mouse brains when compared to the FDA-approved MC3 LNPs following intravenous administration. This approach resulted in robust transfection of both neurons and astrocytes across multiple brain regions. In addition, MK16 BLNPs have successfully demonstrated the ability to deliver mRNA to human brain explants ex vivo. These results indicate potential for clinical applications, alongside a favorable tolerability profile across multiple doses [28].

4.4. Ionic liquids

Ionic liquids are the novel class of biomaterials, with tunable physiochemical properties that have recently emerged and lead to a substantial progression in the delivery of therapeutics and traditionally barrier-hindered molecules through a versatile range of administration routes (transdermal, buccal, intranasal, and oral) [244]. Ionic liquids are salts composed of bulky and asymmetric anions and cations presenting as viscous liquids below 100 °C. Coating LNPs with Ionic liquids can enhance the accumulation of LNPs at the BBB site and consequently improve brain uptake and delivery across the BBB. For instance, LNPs coated with choline-2-hexenoate as Ionic liquids, can cause the therapeutic dose to traverse the BBB through receptor-mediated transcytosis mediated through cation-anion transporter-mediated transcytosis and choline transporter-like proteins 1and 2/monocarboxylate transporter [244,245].

As well, 2-hexenoic acid anion can engage with anion transporters at the BBB to facilitate the Ionic liquid-coated LNPs transportation across it. Anion transporters such as monocarboxylate transporters are expressed on the BBB and with potential capability to interact with the 2-hexenoic acid (Ionic liquids component) to assist in their transport across the BBB [236,246,247].

Coating of LNPs with Ionic liquids can lead to a higher dose at BBB and a lower dose reaching the liver and spleen compared to the uncoated LNPs [230,236].

The intravenous administration of Ionic liquid-coated LNPs facilitates their interaction with RBCs through a shearing-based hitchhiking mechanism. This process involves the shearing of Ionic liquid-LNPs against blood vessel walls and the affinity of the 2-hexenoic acid anion component of the Ionic liquids for sphingomyelin and phosphatidylcholine present on RBC membranes. There is a hypothesis that following intravenous administration, Ionic liquid-coated LNPs may attach to RBCs and be transported within the circulatory system. Given that 15–20 % of the body's blood supply is directed to the brain, a higher dose of Ionic liquid-coated LNPs can reach the BBB in conjunction with RBCs [236,248].

Modifying the surface of LNPs with biocompatible Ionic liquids could also significantly alter their protein binding properties, potentially enabling their redirection to distant targets like the BBB. Therefore, coating LNPs with Ionic liquids might enhance their 'stealth' characteristics compared to conventional LNPs, leading to greater accumulation in the brain endothelial cells that line the BBB. It has also been shown that mouse brain endothelial cells and motor neurons internalize Ionic liquid-coated LNPs and Ionic liquid-incorporated LNPs to a higher extent than conventional LNPs. The surface characteristics of the Ionic liquid-coating LNPs seemed to affect the extent of uptake by the brain endothelial cells [236,244].

4.5. Combination Therapies: Anti-inflammatory drugs

Nanoparticle encapsulation offers the benefit of improving the targeted delivery of immunosuppressants to specific inflamed tissues, while also minimizing their systemic distribution, which can have toxic effects [249,250]. For instance, extensive research has been conducted on dexamethasone and rapamycin-loaded nanoparticles, which have shown promise in alleviating inflammation and inducing antigen-specific tolerance in various diseases including MS [21,250,251]. Additionally, LNPs containing dexamethasone have demonstrated the ability to enhance mRNA delivery compared to LNPs without dexamethasone, while also reducing levels of inflammatory cytokines in mice. Building upon this concept, various other nanomedicine-based strategies could be employed to deliver alternative anti-inflammatory drugs and other corticosteroids, in a more controlled manner. Consequently, LNPs can be deliberately designed based on the selected biomaterial and encapsulated cargo to evade or counteract the immune system [21].

4.6. Enhancing brain specific delivery

Microbubble-assisted focused ultrasound (FUS)-induced BBB opening is a novel technique that enables brain drug delivery in a minimally invasive manner. After intravenous injection of ultrasound-responding microbubbles, FUS application generates oscillation or cavitation energy, opening the BBB temporary and locally for selective drug delivery [17,30].

The combination of low-power ultrasound and intravenous administration of preformed microbubble contrast agents results in mechanical stimulation of the blood vessel via stable cavitation when the microbubbles interact with the ultrasound field, causing temporary and repeatable opening of the BBB. The extent of BBB opening induced by FUS can be influenced by factors such as the ultrasound frequency (and consequently the pressure), the duration of the ultrasound pulse, the pulse repetition frequency, as well as the size and dose of the microbubbles [17,30,31,252].

It is worth mentioning that FUS-mediated delivery of various adeno-associated virus serotypes suggested increased neuronal transduction into CNS with higher BBB permeability relative to other regions [253]. Moreover, FUS-induced BBB opening size showed dependency on the acoustic pressure [254]. FUS-mediated delivery enhances BBB permeability, increasing neuronal transduction and bioavailability, however, it can cause off-target effects and the extent of BBB opening depends on acoustic pressure [255].

This technique has shown effectiveness in patients with amyotrophic lateral sclerosis and Alzheimer's disease, and in preclinical studies for delivering anticancer drugs and plasmid DNA to the brain. More stable microbubbles with higher echogenic gas content have achieved BBB opening [17,30]. However, to the best of our knowledge, there have been no reports on the delivery of mRNA-LNP using microbubble-assisted FUS-induced BBB opening.

Magnetic resonance imaging-guided FUS as a non-invasive and disruptive technology can also lead to the delivery of targeted therapeutics through opening BBB for therapeutics of any size. This technique can overcome significant limitations of therapies with spare normal structures at CNS. Magnetic resonance imaging-guided FUS does not have a cumulative effect and is suitable for repeated treatments and controlled real-time management with immediate tissue destruction and enhanced permeabilization [252,256,257].

The BBB can also be temporarily disrupted using mannitol, a hyperosmolar agent. In both rats and humans, intra-arterial injection of mannitol has been shown to induce transient BBB disruption by reducing endothelial integrity, as evidenced by decreased expression of junctional adhesion molecule-1 and occludin. Notably, BBB disruption via mannitol has demonstrated protective effects in hemispheric ischemic stroke [32].

In addition, mannitol-modified poly β-amino ester nanoparticles carrying plasmid DNA combined with systemic mannitol injection to facilitate BBB crossing, have been shown to effectively transfect neuronal cells and enhance transport across BBB endothelial cells through caveolae-mediated transcytosis. This process occurs without triggering neuroinflammation or compromising the integrity of tight junctions, highlighting its potential for safe and efficient gene delivery [33].

Mannitol can also chemically disrupt the BBB following intra-arterial infusion [258]. It induces global BBB disruption by osmotically shrinking endothelial cells, creating fenestrations in cerebral vessels. However, this non-specific disruption results in inconsistent drug delivery (with up to tenfold variability) and an increased risk of off-target effects. While generally safe, the technique lacks precision in therapeutic targeting [258,259].

Numerous clinical trials are investigating the use of mannitol for BBB disruption in brain tumor treatment [260]. Combining this approach with optimized LNP formulations could significantly enhance the efficacy of brain-targeted mRNA therapies.

4.7. Alternative routes of administration: A key factor influencing efficacy and safety

To enhance the distribution of mRNA therapeutics through the BBB and achieve optimal functionality within the CNS various methods of administration (as alternative approaches) could be employed. Different methods for CNS delivery include the invasive and non-invasive techniques such as the minimally invasive nasal depot (MIND) technique (for effective nose-to-brain delivery of mRNA) [261], intracerebroventricular (ICV) injection [262], intraparenchymal administration, intrathecal injection [263], or the application of ultrasound [207,215,261,264].

Localized delivery to the brain can be accomplished through ICV injection, intrathecal (lumbar puncture) or intracisternal (intra cisterna magna) administration, and convection-enhanced delivery (CED), which may or may not involve the use of reservoir pumps [252].

Delivering RNA-based therapeutics to the brain remains a major challenge due to the protective nature of the BBB. Even in neurological disorders where the BBB is compromised, due to disrupted tight junctions, altered transcytosis, or increased paracellular permeability, RNA therapeutics still exhibit limited penetration, resulting in low mRNA bioavailability (typically <1–2 %) [207,260,265].

The effectiveness of delivered mRNA is significantly influenced by the chosen route of administration which impacts its bioavailability. Among different routes of administration, ICV delivery can achieve near-complete bioavailability within the CNS, as the therapeutic agent does not need to cross the BBB and is not subject to systemic degradation or clearance. In this regard, ICV delivery of antisense oligonucleotides in in rats and monkeys after targeting superoxide dismutase 1 (SOD1) encoding gene, showed a wide distribution of antisense oligonucleotides within CNS, especially the regions affected in the major NDD. All these processes effectively diminished SOD1 protein and mRNA levels throughout the CNS [266]. Studies show that directly injecting LNPs containing mRNA into the brain can lead to widespread distribution of the mRNA in specific brain regions, such as the striatum and hippocampus, indicating effective delivery. Intracerebral injection ensures that the mRNA is delivered directly to targeted areas, achieving a high local concentration of the therapeutic agent [231].

Intraparenchymal administration bypasses the BBB, however it usually does not provide optimal CNS transduction or biodistribution by limiting transgene expression at the administration site depending on the vector design and RNA construct [267,268].

Alternative routes of administration, including the nose-to-brain approach can offer a rapid onset of action, decreased systemic adverse effects. Intranasal administration offers a non-invasive route, bypassing the BBB and utilizing the olfactory and trigeminal neural pathways to

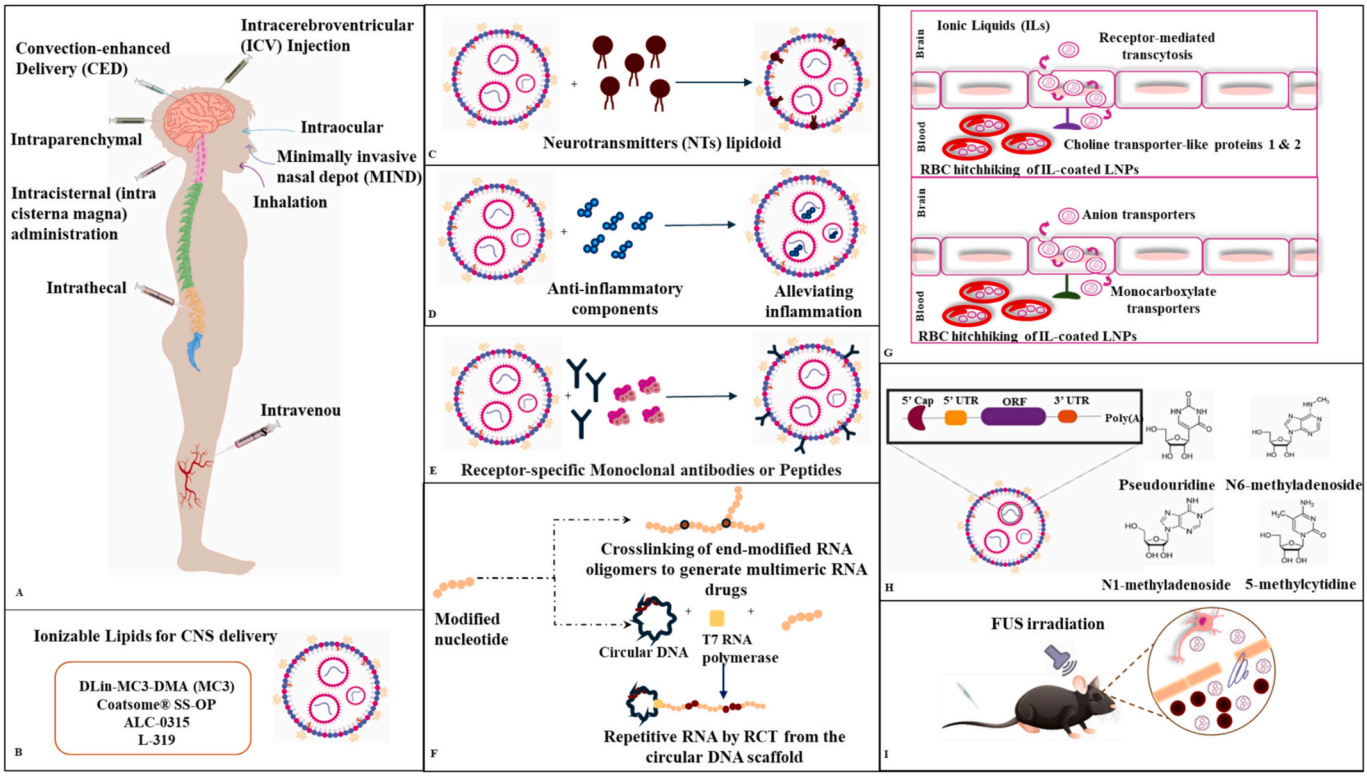


Fig. 9. Schematic illustration of strategies to enhance mRNA-LNP effectiveness. (A) Administration Routes: Utilizing diverse administration routes can optimize the distribution of mRNA therapeutics throughout the body. (B) Certain ionizable lipids, such as Coatsome SS-OP, ALC-0315, L-319, and Dlin-MC_3-DMA have been used for delivery of mRNA-LNP to brain. (C) Neurotransmitter-Derived Lipids: Synthetic lipids derived from neurotransmitters can transport small molecules, nucleic acids, and proteins by capitalizing on the inherent BBB-penetrating capabilities of neurotransmitters. (D) Combination therapy: Combining LNP-mRNA candidate therapeutics with corticosteroids (e.g., dexamethasone) or other small molecules known to inhibit innate immunity may result in improved outcomes (E) Functionalization of LNPs with targeting ligand: Attaching monoclonal antibodies, nanobodies, peptides, proteins to the surface of LNPs can enhance delivery to the brain, offering a more targeted approach. (F) Multimerization through chemical conjugation and enzymatic elongation of RNA therapeutics can bolster the expression of target genes to ensure high stability, high efficacy, and prolonged action of RNA-based drugs (G) Ionic Liquid (IL) surface modification can increase LNP accumulation at the BBB: Coating LNPs with biocompatible ILs can alter their protein-binding properties, potentially redirecting them to distant targets like the BBB and improving accumulation in brain endothelial cells. This modification may also provide stealth characteristics compared to conventional LNPs. (H) Chemical and physical modifications on mRNA could be beneficial for the enhancement of their efficacy. (I) Focused Ultrasound (FUS) with microbubbles: This innovative technique employs microbubble-assisted FUS to temporarily and locally open the blood-brain barrier (BBB), facilitating targeted drug delivery in a minimally invasive manner.

deliver drugs from the nasal cavity to the brain [269]. In this regard, through MIND route, the delivery of a single-stranded oligonucleotide-based compound, named AntagoNATs, and BDNF expression provided >10 % efficiency compared to ICV administration [261].

Regarding systemic administration with modified LNPs, in one study by Mitchell' s group, functionalizing LNPs with specific peptides has been shown to enhance mRNA transfection in the mouse brain while reducing off-target effects in other organs, such as the liver. These LNPs, optimized for factors like serum protein adsorption, intracellular trafficking, and endothelial transcytosis, showed enhancing mRNA delivery to non-neuronal brain endothelial cells and neurons [26]. In addition, as mentioned earlier, recent studies have reported the creation of BBB-Crossing LNPs (BLNPs) that can safely and efficiently deliver mRNA into the brain after intravenous injection, marking a significant advancement in non-invasive brain delivery methods [28].

Nanoparticles hold great potential for overcoming the challenges of low bioavailability and inadequate pharmacokinetic profiles associated with naked RNA, ensuring more effective therapeutic outcomes. Moreover, optimization of the nanoparticle's physicochemical properties and surface modification with different targeting ligand could be applied to increase the efficiency of the nose-to-brain delivery [270]. The most notable and important route of administration is intracerebral injections mostly for brain tissue delivery of mRNA-LNPs in neurological disorders. Intracerebral administrations are clinically feasible and avoids the tissue barrier that CSF or BBB creates with consequent higher brain transfection levels as this route specifically targets local regions of the brain [231,271,272].

Overall, LNPs are considered the suitable choices for the transport of genetic materials to the CNS. The development of LNPs with enhanced stability and targeted delivery capabilities could improve the therapeutic index and reduce the risk of off-target effects, making RNA therapy a more viable option for long-term management of NDD [273]. However, current administration routes often pose challenges for patient compliance, necessitating the development of safer, more efficacious delivery platforms [16]. Therefore, determining the optimal dosing regimen and administration route (e.g., intravenous, intranasal, or direct injection into the CNS) is crucial for achieving therapeutic levels of mRNA in the brain [207].

While various techniques have been explored to enhance the delivery efficiency of RNA therapeutics across the BBB, each has its own set of advantages and limitations. The development of brain-targeted mRNA delivery systems remains a critical area of research to improve the bioavailability and therapeutic efficacy of RNA-based treatments for neurological disorders. Different strategies suggested in order to boost the effectiveness of mRNA-based LNP treatments are summarized in Fig. 9.

5. FDA-approved RNA therapeutics and clinical trials for management of neurological disorders

The specifically mRNA-based LNPs that are FDA-approved are the Moderna vaccine (trade name: SPIKEVAX (elasomeran)) and Pfizer-BioNTech vaccine (trade name: Comirnaty (tozinameran), both of which were approved in 2021 for COVID-19 [274]. The evolving exploration for mRNA platforms drove the need to develop scalable production strategies for mRNA to support pre-clinical studies and early phase clinical trials [275]. The immunogenicity of in vitro transcription (IVT) mRNA can be beneficial for vaccination but may also suppress antigen expression and affect vaccine efficacy. Chemical modifications, such as pseudouridine and N1-methylpseudouridine, can reduce immune sensing of exogenous mRNA translation. IVT mRNA used in COVID-19 vaccines was optimized for translational efficiency and encoded a stabilized pre-fusion spike protein [36,211].

Regarding the neurological and neuromuscular diseases, most of FDA-approved RNA-therapeutics are confined to ASOs and siRNAs. ASOs and siRNAs are mostly administered intrathecal for treatment of the neurological and neuromuscular diseases [252]. Spinraza (Nusinersen), developed by Ionis Pharmaceuticals, is a notable example of a successful treatment for Spinal Muscular Atrophy (SMA) delivered via intrathecal injection. Other investigational ASOs from Ionis, also administered by intrathecal injection, include Tofersen, which targets the SOD1 gene and is currently in phase III clinical trials for amyotrophic lateral sclerosis (ALS) (NCT04972487, NCT04856982) [207,276–278].

These two approved medications are both for management of neurological disorders. Tominersen is another experimental ASO designed to target the huntingtin gene in patients with HD (NCT04000594, NCT03842969, NCT03342053), with demonstration of a notable decrease in HTT protein levels in the cerebrospinal fluid and exhibition of an acceptable safety and tolerability in treated patients. For therapeutics like ASOs that require repeated administration, intracerebroventricular (ICV) injection is not clinically favored due to its invasive nature, although it is commonly used in research studies [279,280]. AMT-130, a synthetic miRNA delivered by adeno-associated virus, is administered via intrastriatal injection using a microcatheter in patients with HD (phase I/II of clinical trial). Currently, Zolgensma is the only FDA-approved adeno-associated virus-based therapy for neurological diseases, which uniquely benefits from the increased permeability of the infant BBB [252,281]. However, Hovakimyan's team developed a vaccine platform called MultiTEP, which targets pathological Aβ and has successfully completed IND enabling studies. This vaccine specifically. Subsequently, patients struggling with AD have enrolled in a trial, and a Phase I clinical trial was initiated [192]. Our investigations indicate that there is currently no FDA-approved mRNA-based LNP formulation available for the treatment of neurological disorders. Limitations and challenges in mRNA delivery by LNPs.

6. Limitations and challenges in mRNA delivery by LNPs

The NDD often involve complex pathophysiological processes and require innovative approaches for effective treatment. There are challenges and limitations associated with delivering mRNA therapeutics across the BBB, endosomal escape, ensuring stability and efficacy, and addressing safety concerns and immune response.

BBB protects the brain from harmful substances, but it also limits the entry of therapeutic agents. For instance, the capillary microvessels within the brain have developed mechanisms to regulate the passage of molecules and cells between the bloodstream and neural tissue, thereby serving as a protective barrier against potentially harmful or infectious substances. BBB exhibits a degree of impermeability due to the presence of tight junctions and adherens junctions that connect capillary endothelial cells, which are mediated by cell adhesion molecules. Additionally, brain endothelial cells have limited alternative transport routes, such as fenestrae, transendothelial channels, and pinocytotic vesicles, and they express elevated levels of active efflux transport proteins, including P-glycoprotein (P-gp, MDR-1 or ABCB1) and breast cancer resistance protein (BCRP, ABCG2). While the BBB is crucial for maintaining brain homeostasis, it poses a considerable challenge for the effective treatment of various neurological disorders [282–284].

LNPs face limitations in their ability to specifically target tissues other than the liver and may also have concerns regarding their long-term safety profile [285–287]. Thus, overcoming extracellular and intracellular barriers and effective endosomal escape and mRNAs release into the cytoplasm are also the hurdles, which could be addressed by various chemical modifications [286,288,289].

Chemical modifications, such as the incorporation of synthetic 5′-cap analogs and 2'-*O*-methylation, can enhance mRNA stability and translational efficiency, while ionizable lipids in LNPs can facilitate endosomal escape by destabilizing the endosomal membrane under acidic conditions. As mentioned, the lipid composition of LNPs also play a crucial role in enhancing mRNA encapsulation release and their ability to cross the BBB [285,290].

mRNA therapy can also provoke an immune response and cause immunogenicity, triggering the innate immune reaction that could diminish the therapy's effectiveness and potentially lead to adverse effects. This can be mitigated by modification of the mRNA's structure or sequence, optimizing the coding sequence, or suppressing the immune system [213]. Notably, the immunostimulatory properties of mRNA-LNP represent a significant hurdle that must be overcome to successfully transition this platform into clinical practice [291].

It is noteworthy that loading capacity of LNPs is a key area for future development, particularly for delivering large mRNA payloads, where challenges in payload distribution, RNA loading, and transient expression limit efficacy [292].

Although the main challenges of systemic mRNA delivery could be addressed by using LNPs, some hurdles and limitations have yet remained to be resolved. Maintaining the integrity of mRNA-mediated LNP formulations over time is a major hurdle, as mRNAs are prone to degradation through hydrolysis and oxidation. LNPs can suffer physical and chemical deterioration, all of which are exacerbated by heat shocks. To address these issues, these formulations are typically kept at temperatures of −20 °C or lower, although for broader accessibility and economic reasons, it's desirable to find alternatives to such extreme cold storage for mRNA-based therapies as freezing and thawing (common in pharmaceutical transport) is not desirable [287,293,294]. Storage and distribution complexities due to the requirement of deep freeze temperatures further complicate clinical translation [43].

Other limitations and challenges in mRNA delivery by LNPs include suboptimal protein expression, and the need for repeated injections due to transient protein expression, which can trigger immune responses and reduce efficacy. LNPs can also induce unexpected adverse side effects, such as inflammation exacerbation in pre-existing inflammatory conditions, and interact with plasma proteins to promote immune responses.

Additionally, the delivery of LNPs to specific cells and tissues is hindered by inadequate knowledge of LNP physicochemical properties and challenges in targeting organs outside the liver [43]. Of note, CNS comprises various cell types (neurons, glial cells), and achieving selective delivery to specific cell types can be challenging. Recent clinical developments highlight the complexity of translating preclinical successes into safe and efficacious human therapies, underscoring the need for developing organ- and cell-specific LNPs via DNA/RNA barcoding that enabling high-throughput screening and precise formulation adjustments [292].

All in all, numerous challenges must still be addressed in order to facilitate the translation of research findings into clinical applications. These challenges manifest at various levels, including those related to formulation, design, and industrial barriers, as well as physiological, immunological, and clinical factors [295,296].

7. Perspective for the future of RNA therapy

Recent advancements in RNA therapy highlight LNPs as a promising vehicle for targeting the CNS in neurological disorder treatment. [207]. Future research will focus on optimizing LNP formulations to enhance BBB penetration, stability, and degradation properties. Cutting-edge techniques like artificial intelligence (AI), machine learning (ML), and high-throughput screening will help refine LNP compositions, improving pharmacokinetics, stability, and targeting efficiency. Additionally, advancements in CAR T-cell engineering, genome editing, and organ/cell-specific targeting will drive the development of precision mRNA-LNP therapies for more effective treatment outcomes.

7.1. Artificial intelligence (AI) and machine learning (ML)

AI-driven methods could help in LNP-mRNA formulation development and also accelerate clinical translation and personalizes therapies.

7.1.1. AI-driven LNP and mRNA sequence optimization

AI and machine learning (ML) are revolutionizing the design and optimization of LNP-based mRNA therapies, enhancing delivery efficiency. By analyzing datasets of lipid ratios and mRNA sequences, AI can predict optimal LNP compositions, streamlining a traditionally time-consuming and resource-intensive experimental process. AI-driven approaches have significantly improved the design of ionizable lipids by predicting their efficacy. Anderson's team developed neural network-based models that analyze extensive lipid structure datasets (over 9000 LNP measurements) to identify candidates with optimal delivery properties [297]. Additionally, AI-driven LNP design can leverage lipid libraries, genetic engineering, and PEGylation or targeting ligand conjugation [298]. Virtual screening further facilitates the rational design of ionizable lipids by predicting key LNP characteristics, such as apparent pKa and mRNA delivery efficiency [299].

A key area of focus in RNA research is predicting RNA structures and modifications, particularly their folding dynamics. Computational RNA structure prediction is crucial due to the high costs of experimental methods. Recent AI advancements have significantly improved the accuracy and efficiency of RNA structure modeling [300,301]. AI-driven mRNA design algorithms, based on computational linguistics, optimize codon usage, structural integrity, immunogenicity, and protein expression [301]. AI can engineer stable mRNA secondary structures, enhancing protein translation. For example, AI models have shown that Brain-Derived Neurotrophic Factor (BDNF) mRNA has greater stability and higher transcription levels, making it a promising therapeutic candidate for Alzheimer's disease [302].

ML also aids in designing LNPs with tailored properties by optimizing lipid composition and functional groups. Deep learning models predict optimal lipid ratios, ionizable lipids, phospholipids, cholesterol, and PEG-lipids, for better mRNA encapsulation. Random forest regression has identified lipid structural features that enhance mRNA expression, enabling more effective LNP formulations [303]. Interestingly, AI-Guided Ionizable Lipid Engineering (AGILE) accelerates LNP formulation development by predicting optimal candidates from vast chemical spaces, streamlining the identification of effective mRNA delivery systems [304].

Integrating machine learning and high-throughput screening in LNP design accelerates the discovery of optimal formulations. These technologies help identify LNPs with the best pharmacokinetic profiles, stability, and targeting efficiency, ensuring precise delivery to neural populations with minimal side effects [305]. A high-throughput screening platform has been developed to evaluate LNP transport across the BBB and assess endothelial cell transfection with mRNA. This approach has successfully identified promising LNP candidates for mRNA delivery, with potential applications in protein replacement and gene editing [306].

The combination of advanced technologies and LNPs paves the way for personalized medicine in NDD, enabling tailored treatments based on individual genetic and molecular profiles [307]. Additionally, an automated high-throughput screening platform has been designed to identify novel ionizable lipids and optimize LNP-mRNA encapsulation. This system ensures precise formulation, allowing direct comparative analysis of hundreds, or even thousands, of innovative LNPs [308].

7.1.2. Clinical aspect: Early detection and monitoring

AI-driven methods enhance early detection, disease monitoring, and treatment optimization in NDD, driving the development of more effective therapies and improving patient care [309]. AI algorithms play a crucial role in early diagnosis, enabling timely interventions and personalized treatment strategies. Advanced image-processing techniques detect subtle structural and functional brain changes, aiding in the identification of disease-specific biomarkers. These biomarkers facilitate early intervention, assess treatment effectiveness, and support drug development and clinical trials [310].

AI also advances precision medicine by optimizing treatment strategies. It accelerates drug discovery by identifying neuroprotective compounds and predicting drug interactions with neuronal targets. Deep learning models analyze genetic data to personalize treatment options. Additionally, AI supports gene therapy by identifying genetic risk factors and predicting patient responses. ML models further refine CRISPR and gene-editing techniques for hereditary NDD [311].

7.2. Gene editing

Since the early 2000s, gene editing technologies like base editors and CRISPR-based systems have revolutionized therapeutic approaches by targeting genetic defects at their root. Gene editing is emerging as a promising strategy for NDD, aiming to correct or mitigate mutations responsible for neuronal degeneration [17]. Preclinical studies have demonstrated CRISPR-Cas9's potential to correct mutations linked to Alzheimer's, Parkinson's, and Huntington's diseases, offering insights into its therapeutic applications [312].

LNPs are gaining attention for co-delivering mRNA alongside gene editing tools to enhance ND treatment [233]. For instance, LNP-based gene editing can target specific cell types, delivering CRISPR-Cas9 components to neurons affected by Alzheimer's or Parkinson's to correct mutations or modulate gene expression, potentially halting disease progression [21,233,313].

CRISPR-Cas9 is the leading gene editing tool due to its RNA-DNA hybridization mechanism, which offers greater precision than protein-DNA interactions [314]. Base editors, developed by fusing catalytically inactive dead Cas9 (dCas9) or nickase Cas9 (a mutated nuclease that creates a nick in only one DNA strand) with a deaminase enzyme that chemically convert single nucleotide bases, enable precise genetic corrections [315]. Prime editing using LNPs involves the encapsulation and packing of prime editors, as a precision technology for genome editing which consist of CRISPR/Cas9 nickase combined with prime editing guide RNA (pegRNA) and a reverse transcriptase [316,317]. Notably, despite the high efficacy of DNA editors like Cas9, base editors, and prime editors, the potential side effects and mechanisms of action highlight the need for alternative therapeutic pathways [318,319].

7.3. Chimeric Antigen Receptor (CAR) T-cell therapy

In the 1990s, The discovery of CD4+ T lymphocytes driving melanoma regression in the 1990s marked a breakthrough in adaptive T cell therapy. This led to advancements in adoptive T cell (ATC) therapy, particularly with Chimeric Antigen Receptor (CAR) T cells [320]. CAR T-cell therapy has revolutionized cancer treatment, especially for hematologic malignancies, and is now being explored for neurological disorders and NDD. Engineered T cells could potentially target pathological protein aggregates in conditions like Alzheimer's disease [321]. However, CAR T-cell therapy faces challenges, including distinguishing

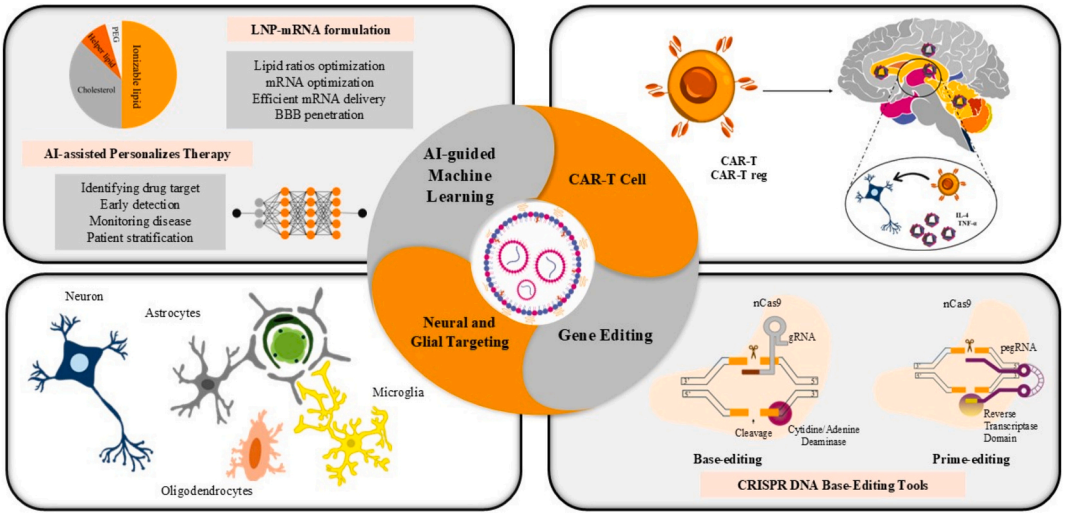


Fig. 10. Perspective for the future of mRNA-LNP in treatment of NDD. AI-guided machine learning is able to accelerate the design, optimization, and clinical translation of LNP-based mRNA therapies for NDD. Advancements in specific cell targeting, gene editing (Base editing and prime editing), and CAR-T cell therapy could transform NDD treatments and enhance clinical applications of LNP-mediated mRNA therapies.

pathological from normal proteins to prevent off-target effects, crossing the BBB, and overcoming the CNS's immunosuppressive environment, which may reduce therapeutic efficacy [322].

CAR T-cell therapy is also being investigated for autoimmune diseases by targeting specific antigens. Identifying new targets and developing innovative CAR-T treatments could transform future therapies [323]. Notably, the first U.S. clinical trials for CAR-T therapy in multiple sclerosis (MS) are underway, with promising potential for halting disease progression in other degenerative autoimmune conditions [324].

CAR T-cell therapy and chimeric autoantibody receptor (CAAR) engineering can mediate B cell depletion in autoimmune, autoantibody-mediated diseases. Additionally, gene editing can knock out T cell membrane proteins such as beta-2 microglobulin, endogenous T cell receptors, and PD-1 [325,326]. Targeting CD19 with CAR T-cell therapy presents a promising, potentially curative approach for immune-mediated neurological diseases. By selectively eliminating autoreactive B cells, it enables long-term remission and reduces the need for continuous immunosuppression [327].

LNPs have demonstrated effectiveness in delivering CAR-mRNA, reducing toxicity, slowing CAR T cell proliferation, and minimizing exhaustion. LNP-mediated mRNA delivery enables transient CAR expression in T cells, allowing targeted therapeutic intervention while reducing the risk of chronic inflammation and off-target effects—critical for precision treatment in neurological conditions [320,328]. LNPs offer a superior alternative to electroporation for mRNA-based CAR T cell engineering, ensuring higher efficiency in mRNA delivery and improved CAR expression [328]. Conjugating antibodies to LNPs enables tunable CAR expression with extrahepatic targeting, leading to significant CAR expression, cytokine release, and up to 90 % B cell depletion [329]. Additionally, spleen-targeted anti-CD3-engineered LNPs have successfully induced CAR T cells against melanoma, enhancing CD8+ T cell cytotoxicity and eliminating tumor cells [330]. LNPs have also been effective in delivering VEGF-A mRNA to various cell types, including epithelial, endothelial, cardiac progenitor, and human umbilical vein endothelial cells [331].

While CAR T-cell therapy has shown remarkable efficacy in relapsed/refractory B-cell lymphomas, its use in CNS malignancies presents unique neurotoxicity risks, such as immune effector cell-associated neurotoxicity syndrome (ICANS) and tumor inflammation-associated neurotoxicity (TIAN) [332]. Notably, a patient in the CARTITUDE-1 trial (NCT03548207) receiving BCMA-targeted CAR T therapy for multiple myeloma developed parkinsonism due to BCMA expression in brain neurons and astrocytes, highlighting the need for neurological monitoring [333]. However, evidence suggests that CNS involvement does not significantly impact therapy severity, supporting its continued use with careful monitoring [334].

mRNA-based LNPs hold potential for managing various diseases and enhancing CAR-T therapy for broader applications [335]. In particular, NP-mediated mRNA delivery offers a promising, safer, and more controlled approach for CAR T-cell engineering, particularly in the treatment of NDD.

7.4. *Targeting specific cell types (Neuronal and glial targeting)*

mRNA-LNPs enhance the targeting of specific neural populations, improving therapeutic efficacy while minimizing off-target effects [245]. In RNA-based therapies for neurological disorders, targeting non-neuronal cells such as glial cells (astrocytes, microglia, and oligodendrocytes), BBB endothelial cells, and peripheral immune cells, can significantly boost treatment outcomes. Microglia regulate neuroinflammation, while astrocytes support neurons by supplying nutrients, maintaining neurotransmitter balance, and preserving the extracellular environment. These cells play key roles in neural homeostasis and neuroprotection [336].

Dual-ligand functionalized LNPs have shown selective uptake by microglia and astrocytes via mannose and adenosine receptor-mediated endocytosis in a traumatic brain injury (TBI) mouse model. Delivering siRNA against p65 using these LNPs silenced p65 in glial cells, reduced pro-inflammatory cytokines, increased anti-inflammatory responses, and mitigated BBB damage [337]. Similarly, adenosine-conjugated LNPs carrying siRNA against TLR4 demonstrated astrocyte-specific uptake near injured brain tissue in a TBI model, promoting neuroprotection and neuronal repair [338]. Additionally, nanoparticles coated with bradykinin B2 receptor antibodies, transferrin receptor, or apolipoprotein E have effectively delivered siRNA and mRNA to astrocytes, further supporting their therapeutic potential [336].

Dahlman's group recently emphasized the importance of quantifying mRNA delivery at the cellular level within the CNS rather than assessing whole-tissue distribution. They developed methods to isolate specific CNS cell types and demonstrated mRNA delivery using liver-detargeted LNPs. Notably, even without targeting ligand, LNPs effectively transfected cerebral endothelial cells in mice following intravenous administration [339]. Beyond neurons, LNPs can be engineered to selectively target glial cells and microglia, highlighting their potential for mRNA

delivery in neurological applications. Targeting diverse CNS cell populations could help modulate the disease microenvironment, enhance neuronal support, and improve therapeutic outcomes. Fig. 10 summarize the perspective for the future of mRNA-LNP in treatment of NDD.

8. Conclusion

LNP-mediated mRNA delivery has demonstrated significant efficacy in preclinical studies, and multiple clinical trials are currently evaluating its therapeutic potential for various NDD. Optimizing LNP formulations, modifying mRNA, and employing BBB penetration techniques (e.g., focused ultrasound, mannitol, etc.) can greatly enhance the effectiveness of mRNA therapeutics. Advanced RNA modalities such as self-amplifying RNA, circular RNA, modified mRNA, and RNA-based gene editing are transforming therapeutic approaches. Additionally, innovations like LNP coating with ionic liquids and targeting ligand conjugation show promise in improving delivery efficiency. By integrating AI, gene editing, targeted delivery, screening technologies, and CAR-T cell therapy, LNPs can be optimized for precise and effective RNA delivery. These advancements could develop clinical applications of LNP-mediated mRNA therapies, ultimately leading to better treatment outcomes for NDD.

CRediT authorship contribution statement

Seyedeh Ghazal Moosavi: Writing – review & editing, Visualization. **Niloufar Rahiman:** Writing – original draft. **Mahmoud Reza Jaafari:** Supervision. **Leila Arabi:** Writing – review & editing, Supervision, Conceptualization.

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Data availability

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